

The Tor Network and Non-State Actors  
An Analysis of the Diffusion of Power in Three Dimensions

Barbara I. S. Calderon

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## **Abstract**

The diffusion of power was a phenomenon first described by Susan Strange in the context of International Political Economy (IPE) in 1996. The phenomenon encompasses the concepts of relational and structural power, which are essential for understanding the IPE discipline. Although described in the 1990s, the diffusion of power did not address the cyber domain, nor explicitly the technological revolutions that culminated in the global computer network. In cyberspace, various actors operate to impose their authority over individuals and services. In this context, at the beginning of the 21st century, the low-latency anonymous network known as "The Onion Router" emerged, being part of the so-called "Dark Web" — a region of cyberspace characterized by active efforts to shield communication, privacy, and specific users. This research aims, through journalistic articles published between 2007 and 2017, to analyze the diffusion of power in the TOR network across three dimensions: authority, control, and results.

**Keywords:** Diffusion of Power. Dark Web. TOR Network. Structural Power. Relational Power. International Political Economy.

# Chapter 1

## Introduction

The present research falls within an area of study still uncommon within the field of International Relations (IR): one that treats the cyber domain as the new stage for actions among various actors in the International System (IS). This area is part of a broader thematic umbrella that discusses the role of Science and Technology (S&T) within the discipline and seeks to understand their influence on international affairs.

Authors have expressed the importance of S&T for the various dynamics addressed by the discipline of IR (SKOLNIKOFF, 1993; WEISS, 2005; KRIGE; BARTH, 2006; MATTHEWS, 1997). Some emphasize that S&T is a powerful and persistent factor, capable of producing social changes and thus culminating in international affairs (SKOLNIKOFF, 1993; MATTHEWS, 1997)<sup>1</sup>. Others point out that this relationship also has an inverse meaning—that is, when S&T themselves are influenced by international affairs, either directly or indirectly (WEISS, 2005, p.297)<sup>2</sup>. That is, just as international affairs are influenced by technological development, the reverse would also be true.

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<sup>1</sup>Indeed, according to Skolnikoff (1993), after the advent of World War II and the discovery of a myriad of scientific breakthroughs—including the discovery of nuclear fission by the U.S. Manhattan Project—any doubt about the relevance of technology for the purposes of the State was extinguished. In this era, almost all the technology that the world may assimilate or deal with is the result of a calculated production of decisions made within existing political processes. In fact, the author himself questions which aspects, if any, of international affairs have not yet been "touched" by S&T. In particular, Skolnikoff (1993) points to evidence of the evolution of S&T in international affairs: the massive deployment of strategic nuclear forces; the Chernobyl nuclear accident; the monetary transactions occurring abroad through digital financial markets—which, at the time, already surpassed 500 billion U.S. dollars daily; and the near-total eradication of smallpox. Another author, Jessica T. Matthews (1997), in the mid-1990s, pointed out the relative decline of States in the face of the rise of non-state actors on the global stage. These actors were to be "catapulted" onto the global stage due to the telecommunications and computer revolution—the political and social consequences of which, according to her, had until then been virtually ignored by academia.

<sup>2</sup>Weiss (2005) points out that the direct effects can be felt in four ways: first, public opinion directly affects public support for investments in S&T; second, the goals of foreign policy have the capacity to alter national priorities, reallocating certain investments to the detriment of others; third, the state of relations between two countries affects the migration of scientists, their communications, collaborations, access to study objects, etc.; fourth, international agreements affect the global strength directed toward the protection of intellectual property. Indirect effects operate through mechanisms of Economics, Law, Politics, and Culture.

Despite the relevance of the topic for IR, the academic literature on it remains limited. In this regard, Eugene Skolnikoff stands out as one of the major figures in the convergence between S&T and IR. His work *The Elusive Transformation: Science, Technology and the Evolution of International Politics* (1993) became a reference point regarding the consequences of scientific and technological changes on the evolution of international politics in various dimensions. In this work, Skolnikoff (1993) also reflects on the reasons why the topic of S&T has been neglected by scholars in IR. He suggests, for instance, that information and knowledge arising from S&T are not treated with the importance they deserve because the perception, on the part of IR scholars, is that they are sometimes “obscure” topics. This obscurity is the reason they are relegated to the second or even third plan.<sup>3</sup> Author Weiss (2005) complements this view of difficulty in assimilation: according to him, the topic of S&T, when addressed by the field of IR, is approached without a defined logical structure. It is as if the subject were composed, apparently, of residual topics from other themes on the IR agenda.

In the 1990s, the revolution of information systems, as an aspect within S&T, was a relevant point of academic discussion. Regarding authors who reflected on the interface between the theme and IR, four scholars stand out: Strange (1988), Skolnikoff (1993), Builder (1993), and Nye (2011). In common, these authors indirectly pointed out how technological changes can impact State power.

In 1988, before the commercialization of the Internet, the British scholar Susan Strange published *States and Markets*—considered a starting point for understanding structural power and, for this reason, a major milestone in the literature of IR and International Political Economy (IPE). In this work, while discussing the *knowledge structure*, Strange (1988) reflected on the technological revolution emerging at the end of the 20th century. In particular, three considerations stand out: (1) the development of sophisticated computer systems with accessibility (availability and low cost) to the masses; (2) the expansion of communication systems using satellites orbiting the planet; (3) and the aspect of language digitization, which significantly reduces the linguistic barriers separating human groups. In general, Strange (1988) outlined insights into the technological revolution and its implications for the provision and control of information and communication systems. She also highlighted changes in the use of language and non-verbal communication channels; in perceptions she deemed fundamental to the human conditions that influence judgments and, through them, political and economic policies and decisions. These points are essential for understanding the dynamics of the knowledge structure, the diffusion of power, and the impact of these elements on State power.

Another scholar, Eugene Skolnikoff, in 1993—during the threshold period of the com-

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<sup>3</sup>Skolnikoff comments on this when he states that “even scholars concerned with theoretical issues in international relations tend to treat science and technology as static ‘givens’, or as emanating from impenetrable black boxes.” (SKOLNIKOFF, 1993, p.9)

mercial and global opening of the worldwide computer network—did not fail to ponder information technologies. Although he did not explicitly address the Internet and its implications for international politics, he discussed the relevance of recent technological changes to political power. According to him, the introduction of information technologies would considerably increase the limitations on the centralization of political power.<sup>4</sup> For Skolnikoff (1993), this conclusion is so widely accepted among commentators on international politics that analyses of international affairs, to a large extent, refer to information technologies as key factors that would undoubtedly bring about changes in the international system. (SKOLNIKOFF, 1993). However, he makes it clear that this will not result in the dissolution of the state or a reduction in its relevance in the international system.

At the threshold of the commercial opening of the Internet, Carl H. Builder was another author who contributed to the analysis of the impact of the information systems revolution on international relations. According to Builder (1993), part of the enthusiasm about the new information age stems from the possibilities that the exploitation of hardware offers in relation to power. This era, responsible for shortening the distances across the globe, disseminated a new perspective on power based on information. For Builder (1993), the state faces a dilemma: should it allow or deny free communication through the new information systems? Permitting it would result in granting the power of information to individuals. And, in possession of this power, they would have sufficient capacity to challenge various hierarchies that were themselves established and maintained through the logic of control and denial of information. On the other hand, denying access to communication and the technologies arising from this new information age could mean missing the opportunity to firmly integrate into the global economy.<sup>5</sup>

Finally, in 2011, the renowned IR scholar Joseph Nye Jr. published the work *The Future of Power*, in which he discussed the relevance of the new information age for rethinking power. In this work, Nye reflects on the technological revolutions of the late 20th century and how they gave rise to a new form of power, which he termed “Cyberpower.” It

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<sup>4</sup>Especially regarding these limitations, he states that “The effects are easier to see and analyze in those nations, for authoritarian governments, aware of the importance of information to their maintenance of power, have made more conscious attempts to control information flow. But, for all governments, information technologies have similar implications for autonomy, openness, and decentralization of power.” (SKOLNIKOFF, 1993, p. 102).

<sup>5</sup>Builder (1993) also addresses other relevant points on the topic. One of his questions revolves around the apparent “revolution” of information at the end of the 20th century: is the world facing a transition or a revolution? If it were a transition, it would be a shift from a society with relative information poverty to one with relative information abundance. As such, this transition would have revolutionary effects on all existing institutions and hierarchies. However, if the world is facing a revolution, the most obvious victims of this dynamic are nation-states and their governments. This does not mean they would disappear, but they would give way to new actors. Thus, an important consequence of this new information age is the emergence of transnational factions. Similar to Strange (1996), Builder (1993) also discusses the power concentrated in the nation-state and its subsequent diffusion to new non-state actors.



is in this hybrid regime scenario—where the concrete reality of cables, machines, computers, satellites, and other infrastructure merges with the virtual abstraction of digital bytes and interactions among agents—that the fabric of the new domain of the 21st century is woven: the cyberspace. In this realm, the state and new actors emerge to seek space and compete for power. (NYE, 2014). However, he advises caution when considering power in this domain, as cyberspace will not replace geographical space nor extinguish state sovereignty. (NYE, 2010).

In summary, the aforementioned authors, despite making relevant considerations about the ongoing technological changes at the end of the 20th century and their impact on state power, were cautious in asserting that such changes do not signify a reduction in the relevance of the state in the international system (IS), nor its extinction.

Before the commercialization and mass use of the Internet, if there was already discussion about the need to include the topic of Science and Technology (S&T) in the range of issues relevant to the field of International Relations (IR), today this need is paramount.<sup>6</sup> There is a need for part of the research efforts to focus on the intersection between Information Systems, Science and Technology, and International Relations—considering the development of the topic and its significant public interest, especially in the range of issues focused on cyberspace. The latter, in fact, has instilled political and economic implications significant enough to reverberate in research across different fields of study that intersect with themes on the IR agenda.

When it comes to research and investigations in the field of IR and cyberspace, the topics vary widely. Some research focuses on the technological dimension of the Westphalian system in the Internet age (BRUNN, 1998; MILLS, 2012; TIKK-RINGAS, 2012; DEIBERT, 2013; DEMCHAK; DOMBROWSKI, 2013). Other studies seek to understand IR through the distinctive lens of Information Technologies (IT)—whether analyzing the convergence of the "digital age" and the "age of terror" (DERIAN, 2003); or examining the role of the state, private actors, and online commerce in regulating information flows (FARRELL, 2006). Much has also been discussed about digital governance—with particular emphasis on the investigations by Canabarro (2014). Other researchers advocate for a view of IR within the cyber domain to investigate digital governance (VAISHNAV; CHOUCRI, 2013; MUELLER; SCHMIDT, 2013). Additionally, there are those who discuss the role of the worldwide web in diplomatic and foreign policy issues (ROSS, 2011). After all, since politics is an action linked to the activities of actors in a collective context, much has been pondered about the political activities of these actors in the cyber domain (CASTELLS; CARDOSO, 2005; ROCHE; BLAINE, 2014; SIEDLER, 2016). Above all, perhaps the most investigated topics are two: cybersecurity (NISSENBAUM, 2005; NYE, 2011; CHOUCRI; GOLDSMITH, 2012; KALLBERG; THURAISINGHAM, 2012;

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<sup>6</sup>Eugene Skolnikoff (1993) is perhaps the leading figure in this agenda that advocates for the inclusion of technology in the issues of the discipline of International Relations.

KRAMER, 2012; HATHAWAY, 2012; CEPIK; CANABARRO; BORNE, 2014) and cyber warfare (BERENGER, 2006; WALSH; BARBARA, 2006; MICHALSKI; GOW, 2007; CLARKE, 2009; LIBICKI, 2009; CORNISH et al., 2010; LAWSON, 2012; HURWITZ, 2013; JUNIO, 2013; CEPIK; CANABARRO; BORNE, 2015; JAJODIA et al., 2015; SCHREIER, 2015; SCHNEIDER, 2016). There is, above all, a significant gap in the study of the Dark Web and IR, this thesis’ topics.<sup>7</sup>

This research is concerned with the structure of knowledge in the context of the cyber domain. Specifically, it seeks to investigate evidence of the diffusion of power through the low-latency anonymous network *The Onion Router* (TOR)—a component of the Dark Web. The latter is a peculiar part of the global computer network, with its own definition. Therefore, we adopt the field of International Political Economy (IPE), from the perspective of Strange (1988, 1996), as the theoretical framework for this study, as we believe this approach has sufficiently relevant elements to explain the phenomenon of the diffusion of power in the cyber domain, encompassing both the plurality of actors and the digital context in which they operate.

## 1.1 Delimitation of the Theme and Research Question

In her 1996 work, *The Retreat of the State: The Diffusion of Power in the World Economy*, Susan Strange explicitly states that structural power can be observed through an analytical framework that defines “who-gets-what” in global society based on four basic structures: the security structure, the production structure, the financial structure, and the knowledge structure.<sup>8</sup> In summary, the objective of her work is to draw attention to the power that originates from these structures, particularly from non-state actors. Due to this power, such actors have the capacity to alter outcomes in the International System. Structural power, as defined by Strange (1988), is derived from and conditioned by these four structures.

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<sup>7</sup>The SCOPUS database—the largest database of abstracts and citations of peer-reviewed literature, according to its own definition—when the user conducts a search using the keyword “dark web,” restricting the fields to “social sciences,” “decision sciences,” “business, management and accounting,” “economics, econometrics and finance,” “multidisciplinary,” and “arts and humanities,” that is, areas related to IR, the search returns 38 results. When, instead of “dark web,” the keyword “tor network” is used, the result is 24 occurrences. In other words, the total is 62 documents. When restricting the fields to “computer science,” “engineering,” and “mathematics,” using the keyword “dark web,” there are 87 results. The keyword “tor network” provides 165 documents. In total, there are 252 documents—more than four times the results in the social sciences fields.

<sup>8</sup>Within each of these structures, power over others—and also over the mix of values within the system itself—is exercised within and beyond the borders of nation-states by: (1) those who are in a position to threaten or provide security; (2) those who are in a position to refuse or provide credit; (3) those who define what is produced, where, by whom, and under what conditions and terms; (4) those who control access to knowledge and information, as well as those who are in a position to define the nature of knowledge (STRANGE, 1996).

Through this analytical framework on the nature of power, the author managed to distance herself from the traditional currents of the IR discipline that upheld the position of the nation-state as the main actor in the International System—and, according to her, often the only one.<sup>9</sup> Strange (1996), in turn, seeks to argue for the existence and relevance of other actors beyond the State in global politics and economy: international organizations, transnational corporations, transnational professions, multinational companies, etc.

The focus of our research is the analysis of the diffusion of power in the context of the anonymous TOR network, embedded within the cyber domain. Due to the nature of this network, the diffusion of power is based on the fourth structure defined by Strange—the knowledge structure. Regarding power and this structure, Strange (1988) stated:

*[The power derived from the knowledge structure is the one that has been most overlooked and underrated. It [...] comprehends what is believed (and the moral conclusions and principles derived from those beliefs); what is known and perceived as understood; and the channels by which beliefs, ideas and knowledge are communicated—including some people and excluding others. [...] Analysis of the knowledge structure is therefore far less advanced and has far more yawning gaps waiting to be filled, than analysis of other structures. [...] Ordinary people in their everyday wisdom have always recognized that ‘knowledge is power’. But in a rapidly changing global knowledge structure such as we have today it is by no means clear to social scientists who has that power. One trouble is that the power derived from the knowledge structure is often very diffused. (STRANGE, 1988, p.119).]*

In the realm of cyberspace, which emerged at the end of the 20th century, it is possible to observe a multiplicity of actors using the Internet to carry out their activities. In this public Internet—free of restrictions and devoid of technologies that provide users and their communications with privacy and anonymity—concerns about surveillance by companies and government entities are voiced by activists and whistleblowers (LANDAU, 2013; SCHEUERMAN, 2014; BAKIR, 2015; WALSH, MILLER, 2015; MURATA, ADAMS, PALMA, 2017). For this reason, the low-latency TOR network—which seeks to provide privacy and anonymity to users and communications through the global Internet—enters the debate on surveillance and privacy in the digital age as an important communication channel. Through the Dark Web, specifically the TOR network, various actors have carried out political operations that have resonated in the media in recent times and have become known to the general public. To illustrate this point, we cite: the WikiLeaks group led by Julian Assange (WIKILEAKS, 2016); the hacktivist group Anonymous (COLLEMAN, 2011); the transnational organization Reporters Without Borders (REPORTERS WITHOUT BORDERS, 2016); former National Security

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<sup>9</sup>In particular, Realism, in its various strands, views the nation-state as the holder of power in international relations—whether through the conception of relational power or material power.

Agency (NSA) agent Edward Snowden—who uses the TOR network, among other mechanisms, to safeguard his communications and ensure privacy (LEE, 2015); the terrorist group Al-Qaeda (COHEN-ALMAGOR, 2012); the Silk Road online drug marketplace (CHRISTIN, 2012); cryptocurrencies like Bitcoin, widely used in commercial transactions on the Dark Web's anonymous networks (KUHN, 2015); and, to a lesser extent, jihadist terrorists and pedophiles (MOORE; RID, 2016).

Since, according to Strange (1988), the structure of knowledge comprises the channels through which beliefs, ideas, and knowledge are communicated, it is reasonable to investigate the diffusion of power within the TOR network and the actors who operate power through this channel. Thus, the research question was formulated to understand how, and to what extent, the TOR network increases the diffusion of structural power in the 21st century. The aim of this research is to investigate the issue of the diffusion of power in this region of cyberspace, examining evidence of the increase or decrease of power by non-state agents. Such evidence should, above all, be tied to outcomes in the geographical real-world dimension of the international system (IS).

## 1.2 Justifications

There are six main justifications that underpin the choice of the theme and scope of this research: first, the rise of the "cyber" theme as a niche for study in International Relations (IR). Second, the urgency of the topic of anonymous networks in the debate on global digital surveillance perpetrated by governmental and corporate entities. Third, the choice of the structure of knowledge over other structures addressed by Strange (1988)—security, production, and finance—is justified. Fourth, the potential academic contributions of this research to the studies on the diffusion of power in the fields of International Political Economy (IPE) and International Relations (IR). Fifth, the choice of the low-latency anonymous network TOR, the selected journalistic media, and the actors presented in this research are also justified. Lastly, the author's personal motivations for the choices and boundaries set in this research.

First, the rise of the "cyber" theme occurred primarily from the new millennium onward. The end of the last century brought about paradigm shifts in technological, political, and social dimensions. According to Castells and Cardoso (2005), this century was responsible for "catapulting" human life into a new phase in technological terms. The period encompassing the significant changes related to the technological "leap" experienced by society is known in the literature as the "Information Age" or the "Network Society." The latter, by definition, "[It is] the social structure that results from the interaction between the new technological paradigm and the overall social organization" (Ibid., p.3). There is a novel element in this society, which is the "microelectronic foundation of network technology that provides new capabilities for an ancient form of social organization:

in networks" (Ibid., p.4). In other words, humans did use the communication networks prior to the 20th century, which were quite common, but the novelty arises from the nature of the network connections, which are now based on digital technology and have a truly global reach (Ibid., p.4).<sup>10</sup>

It is necessary to recognize that, at the end of the 20th century, a new domain emerged that was not adequately embraced by the field of International Relations (IR)—despite its global reach and social and economic character. However, even though it does not have the same volume of studies as the traditional areas of the discipline, considerable advances have been made, as previously highlighted in this research. One of the reasons why the "cyber" theme has gained more space within the scope of the social sciences—although still relatively smaller than in the natural sciences—was the publication, in June 2013, by the British newspaper *The Guardian*, of a series of reports that became known as the "Snowden Revelations," the content of which reverberated across various countries simultaneously. The detailed information provided by the newspaper came from secret documents leaked by Edward Snowden, an employee of Booz Allen Hamilton, a company providing services to the National Security Agency (NSA) of the United States of America (USA). According to Landau (2013), the documents revealed numerous surveillance and espionage schemes focused on the Internet.<sup>11</sup> From the Snowden Revelations, the academic community published several studies on the subject from different perspectives (Landau, 2013; Chadwick & Collister, 2014; Landau, 2014; Lucas, 2014; Scheuerman, 2014; Toxen, 2014; Bakir, 2015; Branum & Charteris-Black, 2015; Lyon, 2015; Merck, 2015; Nocetti, 2015; Qin, 2015; Salvo & Negro, 2015; Walsh & Miller, 2015; Gürses, Kundnani & Hoboken, 2016; Murata, Adams & Palma, 2017). Authors Castells and Cardoso (2005) state that the Network Society currently constitutes the core of contemporary societies. For this reason, it is imperative that new studies be conducted on the "cyber" theme.

Secondly, issues related to the maintenance of rights and civil liberties that can ensure privacy, consent, and free expression on the Internet gained new dimensions after the massive revelations of digital surveillance perpetrated by the U.S. through the In-

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<sup>10</sup>Castells and Cardoso (2005) make a small caveat: although considered global, this network society is diffuse and presents an element of exclusion by not including the entirety of humanity.

<sup>11</sup>Landau (2013) outlines several of them: the domestic telecommunications metadata collection program of Verizon Business Networks Services—including the "who," "what," and "when" of telephone calls; the NSA's "PRISM" program, which targeted specific Internet communications and data storage of individuals outside the borders of the United States and those with whom they communicated; the extent of cooperation between U.S. private companies and the government in providing user and client data; information on U.S. espionage on Chinese computers; the joint operation between the NSA and the Government Communications Headquarters (GCHQ), the British counterpart to the NSA, to monitor the communications of political leaders attending the 2009 G20 Summit in London; the British mass surveillance scheme; the Internet domestic communications metadata collection scheme, among others. Author Lucas (2014) adds more to this: the "XKeyscore" program, revealed in July 2013; and the data chaining program, "Enterprise Knowledge System."

ternet.<sup>12</sup> After the leak of secret documents, Snowden, living in exile in Russia, began advocating for privacy, encryption, and a robust reform of current surveillance systems (Lee, 2015). Privacy and anonymity on the Internet are consistently and openly defended by the former NSA agent. In this regard, the TOR anonymous network has been elevated by him to the status of the most important privacy-enhancing technology of the current era (Ibid., 2015). Snowden’s political stance is not limited to the desire for privacy for ordinary citizens alone. On the contrary, he advocates for other members of society: from those living under authoritarian regimes to professionals committed to reporting and factual accuracy, including political dissidents, whistleblowers, activists, and more. Although the actions of these individuals take place in the abstract realm of cyberspace, their consequences have implications for concrete reality. This demonstrates the political nature of the TOR communication channel in the 21st century. Understanding its functioning, as well as the actions of the various agents operating within it, allows an examination of how power relations are configured in the digital realm today—and, once again, with implications for concrete reality.

Thirdly, our approach aims to analyze the diffusion of power within the context of the TOR anonymous network. This network, by definition, falls under the structure of knowledge, as it is this structure that deals with “the channels by which beliefs, ideas and knowledge are communicated—including some people and excluding others” (Strange, 1988, p. 119). For this reason, and for the purpose of defining the scope of our research, we will analyze the diffusion of power through this structure. It is therefore important to understand both the focus of our analysis and the origins of structural power—two distinct points. While the focus of this research’s analysis of diffusion of power rests on the structure of knowledge—which we believe is best suited to explain the reality of the TOR network and the actors operating within it—the structural power, as conceived by Strange (1988), remains a product derived from the four primary structures, to varying degrees. Thus, we believe it is possible for the focus of analysis to be on one of the structures without compromising conceptual meanings. Strange herself, at times, focuses her analysis on different structures when addressing the diffusion of power.<sup>13</sup>

Fourthly, we believe that this research contributes to academic studies on the theme of diffusion of the power in two ways: first, by bringing the theme into the cyber context; and second, by operationalizing the concept in a way that aids analyses of diffusion of

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<sup>12</sup>The nonprofit organization Electronic Frontier Foundation (EFF), founded in 1990, is a leading advocate for civil liberties in the digital world and is currently battling the NSA in U.S. courts over the agency’s surveillance of Internet infrastructure (Tummarello, 2016). In court, the EFF argues that the NSA violates the Fourth Amendment of the U.S. Constitution by copying and searching data collected from the Internet backbone within the United States (Maas, 2014).

<sup>13</sup>For example, Strange (1996) focuses on the Italian Mafia within the security structure. The Mafia competes with the Italian state for positions within the security structure. Similarly, she acknowledges that the Mafia’s structural power is derived, to a greater or lesser extent, from all four structures—whether through providing physical protection to members and allies, trading illicit goods and services, offering credit to interested parties, or accessing knowledge and information networks.

power. Strange’s works (1988, 1996) did not provide variables that could at least guide the direction of analyses on diffusion of power—how it occurs and to what extent. We believe that constructing variables that can at least delineate the analysis is relevant. Therefore, we will work with three variables during our investigation of diffusion of power within the context of the TOR low-latency anonymous network: authority, control, and outcomes. The methodology involving these variables will be described in the methodological section of this introduction.

Fifthly, we justify the investigation of the TOR network, the selected journalistic media, and the actors presented in this research. Within the context of the networks that make up the Dark Web, the TOR network is considered the largest in terms of nodes and users, as well as being perceived as having the most sophisticated cryptographic technology (Moore & Rid, 2016, p. 15). Moreover, we aim to use alternative journalistic media in the form of online newspapers to build the database on which we apply the variables described above. The selection of these newspapers was based on the *Top 10 Online Newspapers Worldwide Ranked by Unique Visitors*, published in 2012 by ComScore, a U.S.-based company headquartered in Virginia that conducts various analyses related to the Internet domain (ComScore, 2012). Finally, the agents selected for the analysis of the three variables were those who: (1) could be considered part of the knowledge structure, according to Strange’s (1988) definition; and (2) actors whose operations took place through the TOR anonymous network.

The sixth and final justification relates to the author’s personal motivations. These motivations can be summarized as an affinity for the themes of computing, information systems, and technology in general. During childhood, the author learned programming languages and, for the first time, was able to build a website on the World Wide Web (WWW). This experience led her to pursue studies in Computer Science, which were not completed due to her enrollment in Civil Engineering. Her interest and affinity for the Exact Sciences have been long-standing. Both fields were left behind when she began and completed her undergraduate studies in International Relations at the Federal University of Santa Catarina, earning her degree in February 2015. Her final undergraduate project (TCC), though brief, focused on the “Deep Web” and the “Dark Web.” The TCC was supported by three justifications: the latent interest in Technology and Information Systems, especially the world of the Internet; the activist movements in the cyber domain such as Anonymous, WikiLeaks, the TOR Project, and Edward Snowden’s revelations about global communication surveillance carried out by the Five Eyes members, which had political implications for the international scenario; and, finally, the growing concerns regarding the marginalization of these topics within the field of International Relations (IR). In 2016, she enrolled in the Graduate Program in International Relations at the Federal University of Santa Catarina with the intention of continuing her studies on the Diffusion of Power and the Dark Web. In 2017, she published the book *Deep&Dark Web:*

*The Internet You Know Is Just the Tip of the Iceberg* through Alta Books company.

This research is, above all, an attempt to analyze the forces and movements that occur in the cyber domain and reverberate in international political economy and in the daily lives of millions of people around the world.

## **1.3 Objectives**

This dissertation works with one general objective and five specific objectives.

### **1.3.1 General**

The general objective of the research is to understand how, and to what extent, the TOR network increases the diffusion of structural power in the 21st century. To enable the achievement of this general objective, the following specific objectives are proposed.

### **1.3.2 Specific**

1. To examine the theoretical aspect of structural power that fundamentally diverges from traditional approaches to power in International Relations (IR);
2. To propose the operationalization of the concept of diffusion of power, developed by Strange (1996), through three ordinal qualitative variables;
3. To describe the historical, theoretical, and technological aspects that support and accommodate the anonymous TOR network within the cyberspace universe and distinguish it from other communication channels;
4. To analyze the database—composed of subjects from the structure of knowledge selected through newspaper articles—in order to assign values to the variables representing the three dimensions of power that confirm its diffusion;
5. To evaluate, through the previously described specific objectives, how and to what extent the TOR network increases the diffusion of power in the 21st century.

## **1.4 Methodology**

The methodological starting point of this dissertation is the literature review. Initially, we start with a holistic view of the theme of "Science", "Technology", and "International Relations" and the positioning of this research within the aforementioned theme. Subsequently, we map the discussion on the Internet and International Relations in order to investigate the issues already addressed by scholars in the field regarding the cyber



domain and correlate them with this research (as presented in the introduction of this dissertation). We conducted an extensive search on "Power" and "International Relations" to examine how the theoretical framework related to power is configured as an approach within the discipline. Next, we observed how the element of "Diffusion of Power" is addressed by the theoretical framework and other authors in the field—aiming to outline distinctions and similarities. The theme of this research, Dark Web, is situated in cyberspace, and for this reason, we sought to identify the worldwide computer network as its own domain. To achieve this, we explored the literature on the "Origin of the Internet" in its technical aspect to gather knowledge about its operationalization and functioning. Consequently, we understood from the literature on the Deep Web the different nuances of the World Wide Web, among which the Dark Web is configured as just one of the categories. Finally, for the purpose of understanding the specificity and technological nature of the low-latency anonymous TOR network, we examined the literature on the Dark Web and TOR in order to provide the basic inputs through which we conducted the analysis of diffusion of power.

Below is a table composed of the literature review.

Table 1.1: Literature Review

| <b>Categories</b>                                     | <b>Literature on the subject</b>                                                     |
|-------------------------------------------------------|--------------------------------------------------------------------------------------|
| Science, Technology<br>and International<br>Relations | SKOLNIKOFF, 1993; WEISS, 2005; KRIGE; BARTH, 2006;<br>MATTHEWS, 1997; BUILDER, 1993. |

| Categories                           | Literature on the subject                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Internet and International Relations | <p>           NYE, 2011; LANDAU, 2013; SCHEUERMAN, 2014; BAKIR, 2015; WALSH, MILLER, 2015; MURATA, ADAMS, PALMA, 2017; BRUNN, 1998; DERIAN, 2003; CASTELLS; CARDOSO, 2005; NISSENBAUM, 2005; BERENGER, 2006; WALSH; BARBARA, 2006; FARRELL, 2006; MICHALSKI; GOW, 2007; CLARKE; 2009; LIBICKI, 2009; CORNISH et al, 2010; ROSS, 2011; LAWSON, 2012; CHOUCRI; GOLDSMITH, 2012; KALLBERG; THURAISINGHAM, 2012; KRAMER, 2012; HATHAWAY, 2012; MILLS, 2012; TIKK-RINGAS, 2012; HURWITZ, 2013; JUNIO, 2013; DEIBERT, 2013; DEMCHAK; DOMBROWSKI, 2013; VAISHNAV; CHOUCRI, 2013; MUELLER; SCHMIDT, 2013; ROCHE; BLAINE, 2014; CANABARRO, 2014; CEPIK; CANABARRO; BORNE, 2014; CEPIK; CANABARRO; BORNE, 2015; JAJODIA et al, 2015; SCHREIER, 2015; SCHNEIDER, 2016; SIEDLER, 2016; CHADWICK, COLLISTER, 2014; LANDAU, 2014; LUCAS, 2014; SCHEUERMAN, 2014; TOXEN, 2014; BAKIR, 2015; BRANUM, CHARTERIS-BLACK, 2015; LYON, 2015; MERCK, 2015; NOCETTI, 2015; QIN, 2015; SALVO, NEGRO, 2015; GURSES, KUNDNANI, HOBOKEN, 2016; MURATA, ADAMS, PALMA, 2017; TUMARELLO, 2016; CASTELLS, 2001.         </p> |
| Power and International Relations    | <p>           PORTER, 2013; WRIGHT, 1995; DE JOUVENEL, 1957; BALDWIN, 2013; HENDEL, 1953; WALTZ, 1979; WALT, 1987; MOUL, 1989; CLAUDE, 1989; GUZZINI, 2000; SCHWELLER, 2006; KAUFMANN, LITTLE, WOHLFORTH, 2007; LITTLE, 2007; BROOKS, WOHLFORTH, 2008; WENDT, 1999; MORGENTHAU, 1948; SHEHAN, 1996; POLLARD, 1923; PAUL, 2004; WALTZ, 1990; MEARSHEIMER, 2001; SIMONDS, EMENY, 1937; LASSWELL, KAPLAN, 1950; NAGEL, 1975; DAHL, 1961; BACHRACH, BARATZ, 1962; LUKES, 1974; WENDT, 1992; HOPF, 1998; BARNETT, DUVAL, 2005; GUZZINI, 2007; KEOHANE, NYE, 2011; BALDWIN, 1995;         </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Diffusion of Power                   | BUILDER, 1993; NYE, 2011.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Categories       | Literature on the subject                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Internet Origin  | FINKLEA, 2015; CASTELLS, 2001; NAUGHTON, 1999; BARAN, 1964; LAKSHMAN, MADHOW, 1997; CERUZZI, 2003; ASPRAY, CERUZZI, 2008; BANKS, 2008.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Deep Web         | BERGMAN, 2001; FIDÊNCIO, MONTEIRO, 2013; SHERMAN, PRICE, 2001; BECKET, 2009.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Dark Web and TOR | SHERMAN, PRICE, 2001; FIDÊNCIO, MONTEIRO, 2013; BECKET, 2009; GEHL, 2016; CHERTOFF, SIMON, 2015; DEVINE, EGGER-SIDER, ROJAS, 2015; FINKLEA, 2015; ROCHE, 2016; ZULKARNINE et al, 2016; GHAPPOUR, 2017; MOORE, RID, 2016; DINGLEDINE, MATTHEWSON, 2005; PACHENKO, PIMENIDIS, RENNER, 2008; EDMAN, SYVERSON, 2009; CHAABANE, MANILS, KAAFAR, 2010; HOPPER, VASSERMAN, CHAN-TIN, 2010; LI et al, 2010; LI et al, 2011; ALSABAH, BAUER, GOLDBERG, 2012; MOGHADDAM et al, 2012; EDMAN, SYVERSON, 2009; LOESING, MURDOCH, DINGLEDINE, 2010; ELAHI et al, 2012; MCCOY et al, 2008; DINGLEDINE, MATTHEWSON, SYVERSON, 2007; AKHOONDI, YU, MADHYASTHA, 2012; MITTAL et al, 2011; MOGHADDAM et al, 2012; ALSABAH, BAUER, GOLDBERG, 2012; DUNGHEL et al, 2010; JOHNSON et al, 2013; FIFIELD et al, 2012; EDMAN, SYVERSON, 2008; HORSMAN, 2017; THOMSON, 2017; SCHULZE, 2017; BUCHANAN, 2017; SANDVIK, 2017; BIDDLE et al, 2011. |

The theoretical framework of this dissertation is based on the works of the British scholar Susan Strange, which directly and indirectly address the phenomenon of the diffusion of power: *States and Markets* (1988) and *The Retreat of the State* (1996). From this theoretical framework, the need arose to systematize its content in the form of three distinct and complementary dimensions that make up the phenomenon of the diffusion of power: authority, control, and outcomes. These dimensions are implicitly present in Strange's works—but not as formal dimensions per se; rather, they serve as relevant elements for the construction of the analytical framework through which the author's global political economy vision is solidified. The diffusion of power is a phenomenon that occurs separately in each of these three dimensions.

The element of "authority" is interpreted through the lens of the concept of power. On rare occasions, Strange (1996) uses the term "power" when discussing the phenomenon of the diffusion of power, preferring instead the term "authority." When examining the

term 'authority' in her work, we notice a duality of interpretation attached to it: on some occasions, the term is seen as the institution, entity, or subject that embodies the expectation of power (being authority); on others, it is viewed as synonymous with power, or more precisely, with the ability to affect outcomes in such a way that the subject's preferences are imposed (exercising authority). In the first chapter, we aim to substantiate the reasons that indicate the existence of this duality and provide more detailed considerations about how these "authorities" manifest.

In this dimension, Strange discusses the effects of power arising from the presence or involvement of an authority—without the need for the authority to act, operate, or perform activities. Strange (1996) argues that this conception of power finds its origins in feminist approaches. Within a specific context, the presence of a figure understood as authority can be representative of power—and produces effects that are felt and perceived by other figures within that context, leading to consequences. To illustrate, the author addresses the relations between men and women and the inherent power of the male presence in the social context.<sup>14</sup> Furthermore, the presence of authority can emanate from its peers, recognizing three possible scenarios: that the authority is perceived as stronger, has remained the same, or has weakened.<sup>15</sup> We will discuss these ideas in detail

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<sup>14</sup>Especially regarding this notion that power can act merely through presence—and not solely through actions—Strange offers some considerations about this type of power, which she discusses as being "structural", or rather, as a product of the dynamics of established structures (this includes the feminist approach, because power originates from the structural dynamics established in the social context of the patriarchy). She affirms: "A second general point is that 'power over' need not be confined to outcomes consciously or deliberately sought for. Power can be effectively exercised by 'being there', without intending the creation or exploitation of privilege or the transfer of costs or risks from oneself to others, for instance. This recognition of unconscious power is one contribution that gender studies has surely made to international political economy. Male partners may not wish or intend the control they have over outcomes affecting their female partners. But as many women are acutely aware, the social structures within which the partnership exists will make sure that such power exists. [...] This is where the distinction between what I have called relational power and structural power is relevant. In relations with others, it is much harder to think of power being exercised by one party over another unconsciously, without deliberate intent. But when you think of power in terms of power over structures, it is easier to understand that relations existing within those structures are affected, even though it may be inadvertently. The same is true of the power of United States government agencies over outcomes in the international system." (STRANGE, 1996, p.26)

<sup>15</sup>The perception of peers regarding authority—whether it is increasing, decreasing, or indifferent—is an idea indirectly addressed in Strange's 1988 work. The author provides several examples and I quote: "Without the productive power to supply food and capital goods for the reconstruction of European industry, and without the financial power to offer credits in universally acceptable dollars, the United States could not have exercised the power over the recipients of Marshall Aid that it did. Nor was American structural power based only on dominance of the security structure, the production structure and the financial structure. Its authority was reinforced by the belief outside America that the United States fully intended to use its power to create a better post-war world for others as well as for its own people. [...] Moral authority based on faith in American intentions powerfully reinforced its other sources of structural power. [...] A very different example of the power derived in part from the force of ideas would be that exercised within and beyond Iran after the fall of the Shah by Ayatollah Khomeini and his followers. The idea that the Shah, out of greed and lust for power, had fallen captive not only to a foreign country but to a culture and a materialistic belief system alien and inimical to traditional Islamic values had contributed powerfully to the collapse of his government and his own exile. But the power of the ayatollahs in defending and promoting Islamic virtues would have been constrained if they had not

in Chapter One, which focuses on the dimension of authority.

The "control" dimension appears frequently in Strange's works (1988, 1996) when discussing the operationalization of objects within primary structures. She often states that actors positioned in primary structures have power when they control aspects relevant to influencing outcomes in the economic-political scenario (Ibid., 1988). However, the controller of the object influencing outcomes is not necessarily the authority. In the first chapter, we identify various examples from Strange's work where actors within primary structures control objects deemed relevant for influencing outcomes.

The "outcomes" dimension is fundamental in Strange's literature because it determines the authority and control of actors: altering the status quo indicates power, while its preservation may indicate a lack of power.<sup>16</sup> This dimension encompasses both structural power (indirect) and relational power (direct).

In sum, Strange's works indicate that the diffusion of power operates through the three dimensions of "authority", "control", and "outcomes". These dimensions culminated in three ordinal quantitative variables present in our database, compiled from journalistic articles, which underpin our analysis of diffusion of power.

Our methodology is structured as follows: we will outline the three variables used in the diffusion of power analysis, explain their operationalization and scope, discuss the selection of journalistic articles for our database, identify the actors derived from these articles, and detail the process leading to our analysis of diffusion of power through three dimensions.

The independent variable is the TOR Network and the dependent variable is "diffusion of power". Our hypothesis is that the TOR Network enhances diffusion of power in the cyber domain. This research focuses on "how" this diffusion is possible and "to what extent" it occurs.

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also gained control over the state and the armed forces sufficient to confirm their authority both within the country and beyond. Undoubtedly, the power of ideas was indispensable but it could only be used to affect outcomes in conjunction with military capability and economic resources." (STRANGE, 1988, p.32) I argue this excerpt highlights how moral and ideological authority can reinforce structural power, and how peer perception plays a crucial role in legitimizing and maintaining that authority.

<sup>16</sup>We argue that the absence of change does not always indicate an absence of power. Denying access to information, for example, demonstrates power by the actor who withholds it. Strange (1988) highlighted this concept as "negarchy", referring to power derived from resisting changes to the status quo. A classic example is the system of "checks and balances" that operates within the legislative, executive, and judicial branches of government. The ability of one branch to prevent excessive influence from another is indicative of its power. Another example from this research is the Tor Project, which exercises power by resisting digital surveillance over users and messages within the anonymous TOR network. This exercise of power affects outcomes by preventing surveillance, but it does not alter the status quo, as the status quo is the absence of surveillance. A change in the status quo would mean a new reality where digital surveillance is effectively conducted.

### 1.4.1 Variables "authority", "control" and "outcomes"

It is known that when discussing the diffusion of power, Strange (1996) did not use variables that could substantiate the phenomenon being studied. Her methodology, identified by her as "functionalist", examines the various functions of authority in political economy and raises questions about "who", or "what", would be exercising these functions or responsibilities—and, furthermore, with what effects on the outcomes (STRANGE, 1996, p.42). Although she did not approach the phenomenon from the perspective of operationalization, we believe that constructing ordinal quantitative variables is an important attempt to aid in the interpretation of the phenomenon. These variables are able to, at the very least, guide our analytical approach in relation to the diffusion of power.

Based on the reading of her work, our understanding of the diffusion of power is as follows: diffusion of power occurs when an actor has sufficient authority within a primary structure, at either the local or global level, to control an element that is capable of affecting the outcomes. In other words, their control over this element is decisive in determining the outcomes.<sup>17</sup> According to Strange (1996), the diffusion of power helps explain the rise of market power over the nation-state—or more precisely, the rise of the power of non-state actors who operate within each of the primary structures, and sometimes at the intersection of them (secondary structures). Her approach underscores the relevance and strength of non-state actors in the international system and highlights the urgency of addressing them in academic research to better understand international practice. For this reason, she distances herself from the discipline of International Relations—which is almost exclusively concerned with the state actor—and inaugurates International Political Economy, which encompasses non-state actors operating transnationally within the global political economy.

The table below systematically presents the three variables adopted in this research (first column), their scope of action (second column), and plausible values (third column).

Table 1.2: The Diffusion of Power's Variables

| Variables | Scope of Action                   | Ratings       |               |               |
|-----------|-----------------------------------|---------------|---------------|---------------|
|           |                                   | Decrease (-1) | Stability (0) | Increase (+1) |
| Authority | Trust of peers and stakeholders   | Decrease (-1) | Stability (0) | Increase (+1) |
| Control   | Exercise/operation of the element | None (0)      | Partial (+1)  | Absolute (+2) |
| Outcomes  | Impact on the status-quo          | No change (0) | Change (+1)   | -             |

Source: Author.

The first column, "variables," identifies the three variables of the diffusion of power

<sup>17</sup>For instance, she points out that the State holds control over the Armed Forces, which at times act in a way that influences the outcomes so that the preferences of the State take precedence over others. The control over the Armed Forces (and other structures) underpins the State's supreme authority within the security structure. And although the State faces rivals (for example, organized crime—sometimes referred to as "the parallel state") within the security structure, it still remains within this structure as the leading authority. (STRANGE, 1988).

phenomenon: authority, control, and outcomes.<sup>18</sup> The second column, “scope of action,” presents the meaning or the underlying idea behind each variable, i.e., the scope of action of the variable. The third column, “ratings”, represents through maximum and minimum values all the possibilities encompassed by each variable. For example: the “authority” variable has as its scope the trust of peers and stakeholders. This trust can decrease, remain stable, or increase. If a decrease is observed, based on the analysis of the journalistic article, the “authority” variable will be assigned the value “-1”. And so on. The ratings for each dimension are based on the works of Susan Strange. Through them, it was noted that in situations where she mentions authority, she often analyzes whether a particular authority is in decline, growing, or stable. At other times, Strange (1988) indicates the importance of controlling resources that can impact results. And on several occasions, she emphasizes the importance of outcomes in determining the authority, control, and power of an actor. The foundation supporting the existence of the variables, scope of action, and ratings presented in this table is explained in Chapter One of this research and is backed by the literature of Susan Strange.

The first variable, “authority,” refers to the sense of trust conferred by peers and stakeholders.<sup>19</sup> We can say that something or someone possesses sufficient authority within a structure when their peers grant them trust, whether formally or informally. In other words, before anything else, we need to identify the agent/actor in question as a local or global authority within at least one of the four primary structures identified by Strange (1988). If this is possible, the explanatory sequence of the variable follows. Otherwise, we cannot associate the actor with the “authority” variable. Let us consider an example.

A dentist has authority over dental matters because their professional council has granted them this authority through membership. They also have authority over these matters because the university from which they graduated conferred upon them a bachelor’s degree, which allows them to enjoy certain legal prerogatives—including this authority. A drug trafficker has authority over drug transactions in a region because their peers, other drug traffickers, or stakeholders, such as consumers of the product, have granted them this trust. While the dentist is a formal authority, the drug trafficker is an informal authority. This “authority” can increase (in the case of a dentist, this happens when they conduct in-depth research on topics in their field, becoming a specialist in a specific subject); decrease (when the same dentist commits serious errors in the practice of their profession, leading to fewer clients as trust in their work diminishes; if the severity of the error is confirmed, they may be expelled from the professional council and have their legal prerogatives suspended); or remain stable (when the dentist does not become

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<sup>18</sup>These variables are representative of the dimensions of the diffusion of power. In turn, these dimensions originate from the works of Strange (1988, 1996), which serve as the theoretical framework for this thesis.

<sup>19</sup>Strange uses the term “reliability”.

a specialist in any specific subject but also does not commit errors in their professional practice—that is, they simply maintain their level of authority in the exercise of their professional activities).

An actor has power when they possess authority over something. This authority may have been granted through formal or informal means. Let us examine two examples to illustrate this. To demonstrate formality, we will use the example of sovereignty in the International System (IS). Sovereignty is a power exercised in the IS because it is formally recognized by the actors that compose it, or rather, by the peers of the state (state actors) and stakeholders (non-state actors). Regarding informality, consider the example of a drug trafficker in a region. The local population grants them authority based on community belief, either because they provide security in exchange for the ability to market their products or because they are excessively violent in the region, and the community submits to them out of fear of retaliation. Similarly, the US dollar is a currency with power that extends beyond the United States' borders because other states informally grant it authority when they use this currency in their commercial and financial transactions.

When the “authority” variable increases, we assign it a value of “+1”; when it decreases, we assign it a value of “-1”; and when it remains stable, we assign it a value of “0,” as there is no change. This variable, therefore, can take integer values ranging from “-1” to “+1” (including the value “zero”)—depending solely on the context of the situation in which the authority in question is found and the analysis derived from this context.

The second variable, “control,” refers to the sense of exercising or operating a specific object relevant to achieving influence over outcomes. Again, we must first identify whether the agent has control over the object in question. If this can be ascertained, the explanatory sequence of the variable follows. Otherwise, we cannot attribute the “control” variable to the agent.

An actor has power when they have control over something relevant to achieving their desired outcomes. This power is the ability to alter the status quo. They may not be a formal or informal authority; however, if they have the capacity to control this relevant object, then they possess power over the outcomes. This would be the case, for example, of a military pilot in command of a combat aircraft. They do not possess authority; they are an instrument of operation for the Armed Forces to which they are subordinate. However, when, while flying over a region, they are ordered to release the bombs stored in the aircraft, this pilot has the choice to launch the bombs or not. At that moment, they have control over the object relevant to influencing the outcomes: the bombs. During this period, this soldier has power. Their actions can alter the status quo (e.g., not launching the bomb) if they so choose.

Something, or someone, only controls an object if they perform or operate actions to



that end—whether directly (themselves) or indirectly (using another person or mechanism). The “control” variable, when present, determines the degree of operation over the object: none, partial, or absolute. When this variable cannot be attributed to the agent in question, we say that the agent has no control over the specific object and assign the value of “none,” which corresponds to “0,” as their control is nonexistent. If the agent has partial control, they will receive the value of “+1.” And if they have absolute control, they will be assigned the value of “+2.” This variable, therefore, can take integer values ranging from “0” to “+2”—depending solely on the degree of control exercised by the agent in question.

The third and final variable, “outcomes,” concerns the sense of “impact on the status quo.” And, when it comes to the cyber domain, it is relevant to consider the “effects on reality”—the domain in which the actors are situated. Relevant outcomes are those that impact the real geographic plane—not the virtuality of the cyber domain. After all, an outcome only has meaning if it affects some aspect of the reality in which the actors are embedded. Any outcome in the cyber plane will only be taken into account if it also impacts the real geographic plane. A virtual game for leisure can have different outcomes for the players involved. For example, Player A might finish in first place. However, this outcome has no impact on the real geographic plane, as it is played for pleasure. If, by chance, the game took place within the context of a national championship, the first-place finish would certainly have some impact on Player A’s real plane: monetary compensation, a prize, or a title.

An actor can also have power when their actions, with effects on the real geographic plane where the other actors are situated, alter or reinforce the status quo.<sup>20</sup> This is the case of a player in virtual games within a championship. When they compete against the world champion, they have neither authority nor control over the game. However, when they change the outcome (by defeating the world champion, for example), they acquire power (now, they themselves are the world champion). Not altering the outcome (the world champion remains the world champion)—that is, reinforcing the status quo—is also a way of allowing their opponent to maintain the power they already possessed.

The “outcomes” variable can indicate a change in the status quo or its lack of change. If it indicates a change, we assign the value “+1” because the status quo has been altered. If it does not indicate a change, we assign the value “0” because the status quo remains unchanged. This variable, therefore, can take integer values ranging from “0” to “+1” (including “zero,” evidently)—depending solely on the analysis of the change in the status quo in question.

It is necessary to point out that our understanding of the operationalization of the

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<sup>20</sup>As mentioned earlier, the power derived from reinforcing the status quo is related to the idea of “negarchy,” or the force that arises from the denial of changes to the status quo. Strange (1996) reflects on “negarchy” in her work.

phenomenon of the diffusion of power does not concern the gain, stability, or loss of **absolute power** but rather the **level of power**. These are different things. When an actor experiences a loss in their level of power, they lose some power. They do not necessarily lose all their power. But to lose some power, they must have some power to begin with. This is why the situation of having no power and, furthermore, losing a level of power cannot occur. There is no negative power. Now, if an actor experiences stability in their level of power, this means there is neither gain nor loss of power. The actor simply maintains the same level of power they previously held. If they had no power, they remain with none. If they had some power, they retain the same “amount” of power. Their level, wherever it may be, remains unchanged. Finally, if there is a gain in the level of power, this indicates that the actor has necessarily gained some power. If they had none, they now have some. If they already had some, they now have even more—their power is reinforced and reaffirmed.

Finally, we need to make two observations regarding the operationalization of the concept adopted in this research. First, we do not propose that this operationalization, along with the variables, serves as a definitive guide to the diffusion of power. However, we believe it can support a perceptive analysis of the gain, loss, or stability of power in the International System (IS), according to the approach of International Political Economy (IPE). The central idea of this operationalization is to track the journalistic articles selected in this research, in which the actors’ actions are carried out via the anonymous TOR network. We hope this operationalization aids in the analysis of each agent: whether they are losing, gaining, or maintaining the same level of power. Each case can only be evaluated individually, and its context must be respected. The assignment of values to the variables allows, for example, statistical correlation analyses, which, although not used in this research, provide inputs for various analyses. We are also aware that the operationalization of the phenomenon should not be a rigid guide on the subject. Second, the assignment of values to each actor will be based on perception—a subjective characteristic. Each actor, derived from the journalistic articles, will be analyzed through the variables discussed here to frame the perception of the diffusion of power in the context of the TOR network. The actors, who comprise the database of this dissertation, will be identified and discussed in the next section.

### 1.4.2 Selection of News

The search for analytical inputs has an investigative documentary nature, as it was conducted using documents preserved within private journalistic institutions. According to Gill (1991), this type of research has advantages as it is considered a “rich and stable source of data”, does not imply high costs, and does not require contact with research subjects. Pádua (1997, p.62) further argues that documentary research is conducted

using contemporary or retrospective documents, considered scientifically authentic (not fraudulent), and has been frequently used in the social sciences.

We utilized alternative media—more precisely, online newspapers—to compose the reality framework on which the analysis of the diffusion of power was conducted. Riffe, Lacy, and Fico (2008) discuss media research and point out that the scientific method corresponding to "the systematic assignment of communication content to categories according to rules, and the analysis of relationships involving those categories using statistical methods" is called quantitative content analysis. This research method involves a series of techniques, including theoretical literature examination and the development of representative coding categories—aimed at reflecting differences between contents. Its application to mass communication content, such as newspapers, is based on the initial premise that the researcher seeks to develop a social science-based approach through empirical observations and measurements. In other words, there is sufficient suspicion to investigate "theoretical traits" in a specific practical context and thus propose explanations or relationships between different concepts (RIFFE; LACY; FICO, 2008). Regarding this, Riffe, Lacy, and Fico (2008) provide an example:

If members of an ethnic minority, for example, voice concern that they are underrepresented in news media content (in terms of their census numbers), a researcher may propose that racism is at work or that members of the ethnic minority are underrepresented in those occupational groups that serve more often as news sources in the news. Each of these interpretations or explanations involves different concepts that can be “operationalized” into measurement procedures, and each can be tested empirically, as researchers did in the content analyses highlighted previously. (RIFFE; LACY; FICO, 2008)

The present research involves the operationalization of concepts and empirical analysis through the documentary investigation of journalistic articles and categorical measurements—therefore, it employs the scientific method of quantitative content analysis as presented by the aforementioned authors.

Our investigation was limited to the ten most-visited English-language online newspapers, as presented in the justification section. (COMSCORE, 2012).

The British newspaper *Mail Online* is the most accessed by users worldwide, closely followed by the American *The New York Times* and, again, a British newspaper, *The Guardian*. Two Chinese newspapers (*People’s Daily Online* and *Xinhua News Agency*) and one Indian newspaper (*Tribune Newspaper*) are among the ten largest newspapers by user clicks worldwide—although the list is dominated by Western newspapers<sup>21</sup>. According to the ComScore (2012) report, *Advance Digital* is listed in the last position. Despite

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<sup>21</sup>It was not possible to obtain any results for the term “Tor Network” in the newspaper *Xinhua News*

Table 1.3: **The Ten Largest Newspapers in the World by Clicks per User (in Millions)**

| Rank                                                                             | Country | Newspaper                       | Unique Visitors/Month (millions)                                              |
|----------------------------------------------------------------------------------|---------|---------------------------------|-------------------------------------------------------------------------------|
| 1st                                                                              | UK      | Mail Online                     | 50.067                                                                        |
| 2nd                                                                              | USA     | The NY Times                    | 48.695                                                                        |
| 3rd                                                                              | UK      | The Guardian                    | 38.931                                                                        |
| 4th                                                                              | India   | Tribune Newspapers              | 35.862                                                                        |
| 5th                                                                              | China   | People's Daily Online           | 33.026                                                                        |
| 6th                                                                              | UK      | Telegraph Media Group           | 30.083                                                                        |
| 7th                                                                              | China   | Xinhua News Agency*             | 29.987                                                                        |
| 8th                                                                              | USA     | Washington Post                 | 26.007                                                                        |
| 9th                                                                              | USA     | Hearst Newspapers               | 24.174                                                                        |
| <b>10th - Advance Digital (Group of 13 Newspapers) - 22.340 million visitors</b> |         |                                 |                                                                               |
|                                                                                  | USA     | Alabama Local News              | <a href="http://www.al.com">http://www.al.com</a>                             |
|                                                                                  | USA     | Cleveland OH                    | <a href="http://www.cleveland.com">http://www.cleveland.com</a>               |
|                                                                                  | USA     | Mississippi and Gulf Coast      | <a href="http://www.gulfive.com">http://www.gulfive.com</a>                   |
|                                                                                  | USA     | Le High Valley                  | <a href="http://www.lehighvalleylive.com">http://www.lehighvalleylive.com</a> |
|                                                                                  | USA     | Mardi Gras                      | <a href="http://www.mardigras.com">http://www.mardigras.com</a>               |
|                                                                                  | USA     | Massachusetts Local News        | <a href="http://www.masslive.com">http://www.masslive.com</a>                 |
|                                                                                  | USA     | Michigan Local News             | <a href="http://www.mlive.com">http://www.mlive.com</a>                       |
|                                                                                  | USA     | New Jersey Local News           | <a href="http://www.nj.com">http://www.nj.com</a>                             |
|                                                                                  | USA     | New Orleans, LA Local News      | <a href="http://www.nola.com">http://www.nola.com</a>                         |
|                                                                                  | USA     | Oregon Local News               | <a href="http://www.oregonlive.com">http://www.oregonlive.com</a>             |
|                                                                                  | USA     | Central Pennsylvania Local News | <a href="http://www.pennlive.com">http://www.pennlive.com</a>                 |
|                                                                                  | USA     | Staten Island NY Local News     | <a href="http://www.silive.com">http://www.silive.com</a>                     |
|                                                                                  | USA     | Syracuse NY Local News          | <a href="http://www.syracuse.com">http://www.syracuse.com</a>                 |

**Source:** comScore MMX, Worldwide, Age 15+, Oct 2012.

\*Chinese News Agency.

being classified as a newspaper, it comprises a conglomerate of thirteen American newspapers. In total, the ranking of the ten largest newspapers in the world by user clicks, compiled by ComScore (2012), consists of 22 newspapers with active URLs.

To build the database for this research, we accessed the “archives” section (a publicly available database) of each of the listed newspapers and searched for the expression “tor network” (in quotation marks). With the exception of China’s official news agency (*Xinhua News Agency*), all other newspapers returned a list of search results.<sup>22</sup> In total, 302 newspaper articles appeared as results for the search term “tor network.” Of these,

*Agency*, the official news agency of the Chinese government. The search returned some page display issues, appearing to enter a “loop”—when the user makes a request to access the server unsuccessfully, and the procedure repeats indefinitely.

<sup>22</sup>Instead of returning results for the search of the cited term, the newspaper "Xinhua News Agency" repeatedly attempted to access the server but failed to display the results page. For this reason, it was not possible to identify the number of results for the search term "tor network" in the news archives.

only 195 actually dealt with the TOR anonymous network—190 were original articles, while 5 were duplicates.<sup>23</sup> In summary, out of the initial 302 newspaper articles, only 125 simultaneously addressed topics related to the knowledge structure.

Thus, we sought non-state actors who had some degree of social recognition regarding the possession of information and/or knowledge (Strange does not differentiate between these two terms by definition) that were responsible, in some way, for storing information, or controlled the channels through which this information was communicated. Within the universe of 190 original articles related to the TOR anonymous network from the ten largest newspapers by user clicks worldwide, we found that 125 articles discussed non-state actors who met the aforementioned definitions. In some articles, more than one non-state actor was mentioned—so we opted to call “occurrences” the situations involving a single non-state actor and “article” the journalistic reports that could discuss one or more actors. In 14 articles, we identified two actors. Each actor was treated as a unique occurrence deserving analysis.<sup>24</sup> In total, we identified 139 occurrences eligible for analysis—these became the 139 rows in the database for this research. In other words, each row in the database represents a single occurrence involving only one non-state actor positioned within the knowledge structure. Through the newspaper articles, we observed that these actors carried out specific actions via the TOR anonymous network.

Table 1.4: **Number of Occurrences for Analysis (2007-2017)**

| <b>Occurrences*</b>                          | <b>N</b>                    |
|----------------------------------------------|-----------------------------|
| Results for "Tor Network"                    | 302                         |
| Content Unrelated to "Tor Network"           | 105                         |
| Content Related to "Tor Network"             | 195                         |
| Originals                                    | 190                         |
| Actions via Other Means                      | 15                          |
| Actions via Tor Network                      | 177                         |
| Content Unrelated to the Knowledge Structure | 52                          |
| Content Related to the Knowledge Structure** | 125                         |
| Content with 2 Actors per Article            | 14                          |
| Content with 1 Actor per Article             | 111                         |
| Repeated                                     | 5                           |
| <b>Total</b>                                 | $111 + (14 + 14) = 139$ 139 |

**Source:** Author.

\*Situations extracted from newspaper articles composed of a single actor.

\*\*Includes articles composed of actors who are part of the knowledge structure.

<sup>23</sup>The search, which was definitively used in the research, was conducted in the second half of February 2018. The first search was conducted in May 2017.

<sup>24</sup>Therefore, the focus of our research was on the number of occurrences — not the number of articles. For detailed information, see “Appendix 8 — Articles and Occurrences for Analysis by Year.”

Defined by Strange (1988), the knowledge structure encompasses belief, established knowledge, and the communication channels through which beliefs, knowledge, and information are conveyed. These are the three aspects of the knowledge structure. This structure also determines what knowledge is discovered, how it is stored, and who communicates this knowledge—using which means, to whom, and under what conditions (STRANGE, 1988).

The anonymous TOR network is a communication channel through which beliefs, knowledge, and information are securely transmitted. Moreover, the actors originating from the 125 occurrences are embedded within the knowledge structure because they engage in actions related to the discovery, storage, and communication of information and knowledge through the TOR network. According to Strange (1988), actors positioned within the knowledge structure occupy decision-making roles. On this, she observes:

[...] Power and authority are conferred [...] on those who are acknowledged by society to be possessed of the 'right', desirable knowledge and engaged in acquisition of more of it, and on those entrusted with its storage, and on those controlling in any way the channels by which knowledge, or information, is communicated. [...] More than other structures, the power derived from the knowledge structure comes less from coercive power and more from consent, authority being conferred voluntarily on the basis of shared belief systems and the acknowledgment of the importance to the individual and to society of the particular form taken by the knowledge — and therefore of the importance of the person having the knowledge and access or control over the means by which it is stored and communicated. (STRANGE, 1988, p.121).

Actors within the knowledge structure have the capacity to influence outcomes through structural power, which, within this structure, operates less coercively and more through consent. Using this framework—journalistic articles related to the anonymous TOR network and actors positioned within the knowledge structure—we selected occurrences to construct the database. It is from this database that we analyze the diffusion of power across three dimensions: “authority,” “control,” and “outcomes.”

The column “COUNTRY” displays the name of the country where the newspaper is based; “NEWSPAPER,” the name of the online newspaper; “ARTICLE TITLE,” the title of the examined article; “AUTHOR,” the name of the author (or editor) of the newspaper article; “YEAR,” the month and year of publication in journalistic media; “ACTOR,” the actor or subject derived from the newspaper article and through which variables are examined; “CATEGORY,” the thematic group in which the subject is categorized; “PEERS,” the actor’s peers and stakeholders; “AUTHORITY,” the degree of authority of the subject in the given newspaper article;<sup>25</sup> “OBJECT,” the object controlled by the

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<sup>25</sup>The degrees of authority are: increasing, stable, and decreasing. The corresponding values are:

actor according to the article; “CONTROL,” the degree of control the subject has over the object in the article;<sup>26</sup> “STATUS-QUO,” the status quo before the actor’s intervention; “EFFECT ON REALITY,” whether the actor’s actions in the article had real-world effects; and “OUTCOMES,” the degree of change in the status quo in the real world caused by the subject in the given newspaper article.<sup>27</sup>

Table 1.5: **Database Columns for This Research**

| Column            | Information                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COUNTRY           | Country where the newspaper is located.                                                                                                                                    |
| NEWSPAPER         | Name of the newspaper responsible for the article.                                                                                                                         |
| ARTICLE TITLE     | Title of the article.                                                                                                                                                      |
| AUTHOR            | Author of the article.                                                                                                                                                     |
| YEAR              | Year of article publication.                                                                                                                                               |
| ACTOR             | Actor whose actions are described in the article.                                                                                                                          |
| CATEGORY          | Thematic group of the occurrence.                                                                                                                                          |
| PEERS             | Identification of peers and interested parties.                                                                                                                            |
| AUTHORITY         | Identification of the "AUTHORITY" variable in the occurrence. Receives values: "-1" for decreasing authority, "0" for stable authority, and "+1" for increasing authority. |
| OBJECT            | Control object of the actor identified in the occurrence.                                                                                                                  |
| CONTROL           | Identification of the "CONTROL" variable in the occurrence. Receives values: "0" for no control, "1" for partial control, and "2" for absolute control.                    |
| STATUS-QUO        | Presents the status-quo of the occurrence.                                                                                                                                 |
| EFFECT ON REALITY | Identifies whether the actor’s actions have an effect on reality. If positive, assigns "YES". If negative, assigns "NO".                                                   |
| OUTCOMES          | Identification of the "OUTCOMES" variable in the occurrence. Receives values "0" for no change in status-quo and "1" for change in status-quo.                             |
| REPEATED ARTICLES | "YES" to indicate that the article from which the occurrence was extracted is present in another row of the database, and "NO" to indicate the article’s originality.      |

**Source:** Author.

The "CATEGORY" column was established by us and formulated to understand actors in groups of similar themes according to the journalistic articles. There are six categories:

- **Digital Security:** involves themes related to digital security.

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“+1”, “0”, “-1”

<sup>26</sup>The degree of control can vary between “none,” “partial,” and “total,” with the corresponding values being “0,” “1,” and “2.”

<sup>27</sup>The degree of outcomes varies between “change” and “no change” in the status quo. The corresponding values are “+1” and “0.”

- **Surveillance Circumvention:** refers to themes related to shielding against digital surveillance.
- **Journalism & Whistleblowing:** relates to themes of journalism profession and whistleblowing.
- **Online Marketplace:** digital markets operating on the anonymous network.
- **Privacy:** involves news specifically about user data.

In total, we identified 25 non-state actors, which we categorized into the six aforementioned groups:

Table 1.6: Non-state Actors by Category

| Category                    | Actors                                                                                                              |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------|
| Digital Security            | Dan Egerstad, Iranian Cyber Army, Freedom Hosting, Onion Ransomware, SimpleLocker Android Malware, Ransomware       |
| Surveillance Circumvention  | Tor Project, Facebook                                                                                               |
| Journalism & Whistleblowing | Chelsea Manning, Edward Snowden, Harold T. Martin, WikiLeaks, X-Net Group, ProPublica, Strongbox, SecureDrop System |
| Online Marketplace          | Silk Road, Silk Road 2.0, Silk Road 3.0, Alphabay, Evolution, Sheep Marketplace, Farmer's Market                    |
| Privacy                     | Doxbin                                                                                                              |

**Source:** Author.

From the newspaper articles, we extracted the necessary information to populate the database. Besides creating the categories composed of actors with similar themes, we also identified who were the peers and stakeholders interested in each actor's actions and activities. The "PEERS" column aims to identify the subjects whose degree of trust in relation to the actor increases, decreases, or remains stable. As we've seen, this trust degree is translated by the "authority" variable - hence the creation of the "PEERS" column, to provide basis for the "authority" variable. Additionally, we identified the relevant objects for achieving the intended results through the "OBJECT" column. The degree of control over the object exposed in this column translates into the "control" variable. Therefore, it was necessary to identify the control object.

The "STATUS-QUO" column seeks to identify the state of the situation prior to the subject's actions because it is from the change or maintenance of this state that we assign values to the "outcomes" variable. However, before assigning this value, we need to know whether the change or maintenance of the status-quo affects geographical reality, since



this research operates within cyberspace. If there is an effect on reality, we assign the value consistent with the change or maintenance of the status-quo. If there is no effect on reality, we assign the value "zero". Again, the filling of these fields is done based on the content of the newspaper articles.

In summary, the methodology used in this research employs both qualitative and quantitative approaches to analyze the specific objectives. The qualitative approach consists of examination of literature review; preparation and transcription of the database from alternative online newspaper sources; pre-analysis with floating readings; categorization; presentation of results.

The quantitative approach, on the other hand, focused on assigning numerical values to assist in interpreting the power diffusion in each dimension across the selected set of journalistic articles. The general methodological components were: inclusion of online newspapers for database structure; delimitation of the knowledge structure - excluding analysis of other structures presented by Strange (1988); definition of variables and their possible assigned values; period and form of data collection that composed the database.

For data analysis within the quantitative approach, the results are presented in tables, charts, graphs and descriptive statistics - through frequency calculations and relative percentages. Microsoft Excel 2010 was used for spreadsheets and calculations. Data analysis was performed through variable tabulation using descriptive statistics - with absolute and simple relative frequency. Using this method, we made considerations about the phenomenon of power diffusion in the context of the TOR anonymous network, a component of the Dark Web.

### 1.4.3 Database Composition for the Research

For comprehension purposes, we randomly selected a newspaper article to demonstrate how each information column was filled. The article is titled *Global Drug Survey findings: more people buying drugs online in the UK*, published on April 14, 2014 by *The Guardian*, based in the United Kingdom and authored by Ami Sedghi.

The article discusses the increase in online drug purchases in the UK and features the actor *Silk Road*, an online marketplace operating on the TOR anonymous network. According to the article, nearly 60% of respondents were aware of the Silk Road marketplace, while about 44% reported having visited the site. The peers and interested parties are private citizens.

Based on this article:

- We assigned the value "+1" (increasing authority) to the authority variable because more than half of respondents knew about the online marketplace, and the article stated that more people were buying drugs online in the UK. We interpreted that trust in the marketplace among its peers and stakeholders was growing.

- The control object of the Silk Road actor is the online marketplace itself through its webpage and Bitcoin payment mechanism. Therefore, for the control variable, we assigned the value "+2" (absolute control) since control over both the marketplace operations and payment mechanism is administered by a single actor - Silk Road itself. This control is neither dependent on nor shared with any other actor.
- We identified that the status-quo of the situation addressed by the mentioned newspaper article is the fixed number of people in the UK who consume drugs through online purchases. The Silk Road actor changes this status-quo by contributing to the increase in the number of people buying drugs online in the UK.
- This status-quo change has an effect on the real world because people are indeed consuming narcotics and illicit substances. For the outcomes variable, we assigned the value "1".
- Finally, the article in question is original - meaning an article with a single actor. There will be no other actor from this same article.

It is worth remembering that when assigning values to the variables, we also identified the peers and interested parties (authority); the relevant object for achieving desired outcomes (control); the status-quo, so we could examine whether there was any change (outcomes). These findings are fully available in the research database provided in the Dissertation Appendix.

## 1.5 Dissertation Structure

This dissertation is organized into three substantive chapters. The first chapter establishes the theoretical framework within International Relations (IR) that will inform the interpretation of results in the third chapter. The presentation progresses from broad conceptual foundations to specific analytical perspectives.

The first chapter comprises three principal sections. The opening section introduces British scholar Susan Strange, a seminal figure in IR and International Political Economy (IPE), providing biographical context and outlining her major contributions. The second section examines the nature of structural power through two classical IR lenses: the relational approach and the national elements approach. This dual analysis substantiates Strange's theoretical innovation regarding power concepts within IR—a departure from classical disciplinary approaches that constitutes the core of her scholarship. The third section presents Strange's framework of diffusion of power, which requires understanding both relational and structural power concepts as they operate within her structural analytical framework for IPE. This framework encompasses the economic and political structures governing people and services. We conclude the chapter with Nye's (2011)

considerations about power diffusion in the cyber domain from *The Future of Power*, complementing Strange's (1996) work which did not address cyberspace.

Chapter two focuses on the technological dimensions of information systems. The initial section discusses the technological foundations and historical development of the Internet. The second section analyzes the WWW architecture, distinguishing between the "Surface Web" and "Deep Web." Building on Sherman and Price's (2001) taxonomy, the third section classifies the "Deep Web" and situates the "Dark Web" within the broader Internet ecosystem. The chapter culminates with an explanation of the technical characteristics that distinguish the Tor network's low-latency architecture from other Dark Web networks. Tor's unique technical specifications enable operational scenarios impossible on conventional web platforms, making this network the foundational infrastructure for the diverse actors examined in this study.

The third chapter presents our analytical database and methodology for examining the diffusion of power through three operational dimensions: "authority," "control," and "outcomes." This tripartite framework operationalizes Strange's (1996) concept of diffusion of power. Our analysis assigns specific valuations to each variable, with final interpretations derived from value frequency distributions across actor groups. The chapter concludes by presenting key analytical findings.

The concluding synthesis integrates three core elements: (1) the theoretical framework, (2) the technological context, and (3) the database analysis employing our operationalized diffusion of power metrics. This integrated approach enables comprehensive examination of power diffusion phenomena within the cyber domain—specifically focusing on the Tor network ecosystem.

## 1.6 Final Considerations

Traditionally, International Relations (IR) emphasized two main approaches to power — relational power and material power.<sup>28</sup> The reflection on the international scenario through an analytical framework revealed a gap in power studies within IR discipline — since these approaches focused primarily on state-centric analyses. According to Strange (1996), this state-centric view is excessively restrictive as it does not correspond to the reality of the international system: transnational corporations and other non-state actors frequently exercise power that affects millions of people's daily lives, yet they remain ignored in academic analyses. As Strange herself noted:

“What the notion of structural power in world politics, society and economy did was to liberate the study of international political economy from the

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<sup>28</sup>Baldwin (2013) refers to these approaches as "relational" and "national elements of power," respectively.

so-called realist tradition in the study of international relations.” (Strange, 1996).

Furthermore, she argued that power diffusion adequately explained the declining quality of state authority while other actors’ authority was rapidly growing.

However, as Palan (1999) observes, Strange cannot be confined to a standard theory-practice dichotomy for two reasons: she was neither theoretical nor “empirical” in the conventional sense. Rather than being theory-oriented, Strange aimed to develop an analytical framework for exploring issues. This explains why *States and Markets* clearly presents the dilemma between theory and “theorizing”, while lacking empirical rigor (Palan, 1999). We believe this same dilemma appears in *The Retreat of the State: The Diffusion of Power in the World Economy* (1996). Despite introducing innovative themes like the accounting profession and the Mafia in global political economy, the author provides neither methodological framework nor empirical evidence to substantiate power diffusion — at least by academic standards. This observation, however, doesn’t diminish Strange’s pioneering work in IPE. As Palan (1999) suggests, perhaps strict empirical rigor would have raised her concerns about detachment from reality. She explicitly invited researchers to develop theoretical and empirical work on non-state authorities and their impact on IPE’s four structures:

“Like plants in nature, theories and explanations grow out of the dirt of observations of reality. The observations may not be ‘scientific’ in the sense that an experiment in chemistry can be objective. But they are not invented either. Getting your hands dirty with the nitty-gritty details of a technology, or with the decision-making processes of corporate strategies, or of ministerial policymaking, is a good way to test the abstractions of theory, and perhaps to develop alternate theory, or modifications of theories. Moreover, if you can illustrate a theory or a hypothesis with reference to a concrete situation, it often serves to explain more clearly the thrust of the ideas. That is part of the point of my rather scrappy descriptions of non-state authorities and how they affect power structures to be found in part II of the book. They may have been chosen somewhat at random, out of personal interest. But they are supposed to illustrate the theoretical propositions laid out in the earlier chapters. It is my sincere hope that these examples will serve to stimulate younger scholars to more innovative work, theoretical and empirical, on non-state authority in the international political economy. They are by way of being a signpost, pointing not along an open well-trodden track but rather into a mysterious forest of the unknown. Just where the path will lead, I am not at all sure. That is the nature of exploration — and its appeal to the mentally adventurous.” (STRANGE, 1996, p.xvi).

Responding to this invitation, this thesis explores studies on knowledge structure to examine non-state authorities and the diffusion of power in cyberspace, specifically within the TOR anonymous network. We pursue this through two approaches: first, by explicating three dimensions of structural power that, while not formally treated as such in Strange's works, are implicit in her writings; second, by operationalizing the diffusion of power concept through these three dimensions to enable empirical analysis. Our objective is to contribute academically to IPE studies.

## Chapter 2

# Theoretical Framework

The objective of this chapter is to present the theoretical foundation within the field of International Political Economy (IPE) that will serve, in the third chapter, as the analytical basis for interpreting the diffusion of power in the context of the anonymous network The Onion Router (TOR), as reported in journalistic documents.

In this first chapter, we begin from a broad perspective of the traditional concept of power in International Relations (IR) and move toward the alternative conception developed by Susan Strange (1988), which she termed *structural power*. This alternative concept is presented with particular attention to the structure of knowledge. Such structuring aims to provide sufficient theoretical foundations for explaining the phenomenon of *diffusion of power*—introduced subsequently as part of this dissertation’s theoretical framework. The phenomenon of *diffusion of power* is grounded in the notion of structural power.

Next, we explore the three dimensions that, according to our interpretation, are essential for understanding the phenomenon of diffusion of power: *authority*, *control*, and *outcomes*. These three dimensions are present in Strange’s works (1988, 1996), although they were not explicitly formulated as such by the author. This section will therefore provide the necessary elements for assessing the existence and scope of power diffusion within the TOR network (which is part of the knowledge structure)—an analysis developed in the third chapter of this dissertation.

Finally, we briefly present what the IR literature on power has argued in the cyber domain, focusing on the concepts of *cyber power* and *power diffusion* developed by Joseph Nye (2011).

In summary, this first chapter is organized around the following thematic topics: a brief biographical introduction to Strange; a mapping of traditional debates on power in IR, aimed at highlighting the theoretical gaps explored by Strange—which led to her break with IR literature and the development of her structural analytical framework in IPE; a presentation of structural power and the structure of knowledge; a discussion of the phenomenon of power diffusion; an exploration of the three dimensions—*authority*,

*control*, and *outcomes*—and their respective connections with the phenomenon of *power diffusion*; and, finally, an examination of the current IR literature on power in the cyber domain.

## 2.1 Biography and Contributions of Susan Strange

Born in 1923 and graduated in Economics from the London School of Economics (LSE) in 1943, Susan Strange—considered by many the most influential figure in British international studies—was almost solely responsible for establishing the field of International Political Economy (IPE) in the twentieth century. She began her career in journalism before moving into international studies. According to Brown (1998), her late entry into academia helped her remain largely unaffected by ego-driven temptations, as the academic world is shaped not only by learning but also by its institutions and prestige.

The erudite Strange was a resounding voice advocating the fusion of two major disciplines, Economics and Politics. She believed this fusion could foster the emergence of a new field within International Relations. On this point, Brown (1999) argues that there are two fundamental reasons that motivated her to advocate for the merging of these areas in the global context:

1. With few exceptions, she often believed that either economists neglected the role of power in global affairs or were overly confident in their abstract and formal economic models of the real world;
2. Most political scientists appeared excessively impressed by military forces and might, and, for this reason, attributed more power to institutions than they actually possessed.

Brown (1999) also emphasizes that any understanding of Strange’s work must begin with her most evident trait: eclecticism. This characteristic forms the foundation of the interdisciplinarity of the IPE discipline itself.<sup>29</sup>

During roughly ten years, from 1965 to 1976, Strange was a full-time researcher at Chatham House. It was during this period that she published her manifesto *International Economics and International Relations: A Case of Mutual Neglect* (1970), in which she advocated a new analytical approach for International Relations, understood as the amalgamation of Economics and Politics, in order to grasp the reality of interactions and forces

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<sup>29</sup>Some of her works include: *Sterling and British Policy: A Political Study of an International Currency in Decline* (1971); *Casino Capitalism* (1986); *States and Markets* (1988); *Rival States, Rival Firms: Competition for World Market Shares* (1991, co-authored with John M. Stopford and John S. Henley); *The Retreat of the State: The Diffusion of Power in the World Economy* (1996); and *Mad Money: When Markets Outgrow Governments* (1998).

in the global context.<sup>30</sup>

In any case, Strange's efforts to make the field of IPE visible and to attract scholars interested in conducting conjunctural and systemic analyses are undeniable. Scholars note that she generated something very close to an alternative theory of International Relations, although she was neither recognized as a theorist nor did she claim this label herself (Palan, 1999; Cohen, 2016). Her studies on power demonstrate her intellectual approach to issues on the agendas of IR and Economics. Cohen (2016) emphasizes that the element of power is central to any explanation of the character and dynamics of the global economy. Tooze and May (2002) consider Strange's analyses of power to constitute the most significant contribution to the field of IPE. The erudite Strange thus stands out as one of the major figures in the disciplines of IR and IPE.

## 2.2 Power According to Traditional Approaches in IR

International Relations inaugurated the field of study whose domain belongs to what has conventionally been called "international politics" (Porter, 2013, p.4). The American political scientist Quincy Wright (1955) defined "international politics" as the effort to influence "major groups in the world so as to advance the purposes of some against the opposition of others."<sup>31</sup>

The political scientist at Princeton University and emeritus professor at Columbia University, Baldwin (2013), argues that all "politics" involve the element of power, even when other elements are discussed. From this, we can infer that international politics also deals with the element of "power"—both from the inception of IR as a discipline in 1919 and in studies of international political events predating that time (such as the

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<sup>30</sup>From the manifesto arose a conference under the auspices of Chatham House, which brought together groups of scholars from various fields in the U.S. and the U.K. in 1972. This conference, in turn, led two years later to the formation of the International Political Economy Group (IPEG), organized by Strange within the British International Studies Association (BISA), also formed through her own initiatives (Brown, 1999).

<sup>31</sup>This explanation has, as a consequence, a significant political connotation that encompasses the international sphere. However, it does not imply that only states are involved in politics. Other actors are also relevant. Political philosopher Bertrand de Jouvenel (1957) suggests that those involved in politics go beyond politicians themselves, party members, or even social movements with explicitly political objectives. He emphasizes two fundamental points: first, an action becomes political when the assistance of others is a necessary condition for an individual to achieve their goal; second, as a consequence, politics occurs when a project requires the support of another's will. In IR, politics is frequently treated as the domain of rulers and high-level government officials. It is in this aspect that Strange differs from the conventional conception of politics in IR. The British scholar assumes, similarly to Bertrand de Jouvenel, that politics involves not only rulers but all those who act based on the will of others, in order to make assessments regarding the political dynamics of power. For this reason, she is able to observe non-state actors as political subjects and include them within the analytical framework she develops to understand IPE. Both state and non-state actors can be examined through this framework, which differs from traditional IR approaches.



Peloponnesian War, widely analyzed within the discipline).<sup>32</sup>

The element of “power” is widely associated with the Realist theoretical current, although it is not exclusive to it. On this point, Wendt (1999) notes that a defining feature of Realism is indeed the assertion that the nature of international politics is shaped by power relations. However, he also emphasizes that this feature is neither unique nor exclusive to Realism.<sup>33</sup>

According to Baldwin (2013), two traditions dominate the study of power analysis in International Relations: the “national elements of power” approach and the “relational power” approach. The former treats power as a resource, while the latter identifies it as a relationship, whether actual or potential. We will examine these two approaches below in order to understand the landscape of power studies that preceded the development of IPE.

### 2.2.1 The National Elements of Power Approach

The “national elements of power” approach conceives power in terms of *resources*, something that an actor can possess. In other words, these resources are treated as if they were power itself (Baldwin, 2013, p. 277). There is no consensus regarding what these resources are, though some interpretations suggest they consist of the measurable elements that make up a nation—such as *territory size*, *population*, and *level of wealth* (Ibid., p. 280).

This notion of power as property is embedded within theoretical currents that discuss the *balance of power* (Ibid., p. 281). According to Wright (1965), the term “balance of power” embodies the idea that changes resulting from relational political power can be *observed and measured* in practice—in other words, power is a resource that can be quantified. This is consistent with the concept of *possession*, *ownership*, or *control*, which lies at the core of the “national elements of power” approach.<sup>34</sup>

There are three theoretical currents that adopt this perspective: the *classical balance*

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<sup>32</sup>The discipline is nearly a century old. Its origin is believed to date back to 1919, with the creation of the first chair for the study of international politics in the world—at the Department of International Politics of the University College of Wales, Aberystwyth (Schmidt apud Porter, 2013). In 2019, the University of Aberystwyth celebrated the centenary of the discipline, noting that the department’s foundation was a “response to the extreme violence of the First World War, in which millions of people from various parts of the world lost their lives. This foundation constituted an intellectual response to a global event with a normative purpose: to understand the different facets of world politics (politics, law, economics, ethics) in order to mitigate organized violence” (Aberystwyth University, 2017, our translation from written portuguese back to english). Over the decades, the discipline progressed academically, promoting successive waves of theoretical phases: idealism, realism, behavioralism, post-behavioralism, pluralism, neorealism, rationalism, post-positivism, and constructivism (Schmidt, 2017).

<sup>33</sup>According to Baldwin (2013), despite debates on “power” in IR failing to produce a consensus, there is a basic conception of power that is most widely accepted: initially formulated by Dahl (1957), it is the idea that actor A causes (or has the ability to cause) actor B to take an action that B would not have undertaken without A’s intervention. This is the conception of relational power.

<sup>34</sup>The term “balance of power” is widely debated in the IR literature, as different authors argue that it is excessively broad (Pollard, 1923; Sheehan, 1996; Baldwin, 2013). Baldwin (2013) points out that Morgenthau himself admitted using the term to convey four different meanings.

of *power theory* proposed by Hans Morgenthau (1948); *neorealism*, also known as *structural realism* or *defensive realism*, developed by Kenneth Waltz (1979); and *offensive realism*, formulated by John Mearsheimer (2001).

## The Classical Balance of Power Theory

Hans Morgenthau is the most prominent exponent of the *classical balance of power theory*.<sup>35</sup> According to Morgenthau (1948), if we wish to determine the power of a nation, we must distinguish its elements into two major groups: those that are *relatively stable* and those that are *subject to constant change*.<sup>36</sup>

From Morgenthau's perspective, it is the responsibility of those in charge of a nation's foreign policy — as well as those who shape public opinion on international affairs — to correctly assess the impact of these factors on the power not only of their own nation but also of others. This directly relates to the idea of the *balance of power*.

Paul (2004, p. 4) observes that the balance of power theory is grounded in the notion that states pursue two main objectives: to *survive independently* and to *accumulate power* within an anarchical system. Without power, they risk becoming subordinate to the will of others, thereby losing security and prosperity. For this reason, anarchy is the driving force that compels states to increase their power: both security and physical survival cannot exist apart from the maximization of power. Within this context, competition for power becomes a logical outcome.

Weaker states, according to Paul (2004), may lose their security — or even cease to exist — when threatened by an alliance of stronger states. Thus, weaker actors tend to form coalitions in order to balance power against stronger ones.

These observations belong to what is known as the *classical balance of power theory*, which, according to Baldwin (2013), regards military strength as the core capacity to prevail in war, since it is discussed within an analytical framework that presupposes armed confrontation among states. Therefore, this theory's meaning is intrinsically tied to the military context.

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<sup>35</sup>Nevertheless, the concept predates the twentieth century. It was already used by the Greek historian *Thucydides* in his attempt to explain the causes of the *Peloponnesian War*, as reported by David Hume (1742) in his essays. In any case, Michael Sheehan (1996) considers the notion of “balance of power” one of the most important ideas in history for two reasons: first, it is a key concept that helps scholars of International Relations understand the recurring patterns of state behavior under the shadow of anarchy; and second, it has served as a guiding principle for many statesmen.

<sup>36</sup>The first group includes factors such as geography and natural resources (e.g., food and raw materials). The second group comprises industrial capacity, military preparedness (including technology, quality of leadership, and the size and capability of the armed forces), population (distribution and demographic trends), national characteristics (intellectual and moral qualities), national morale (the degree of determination with which a nation supports its government's foreign policy in times of both war and peace, including the stability of morale and the quality of society and government), and the quality of diplomacy (which reflects the quality of government, the balance between resources and policy, and popular support). (MORGENTHAU, 1948).

## Neorealism

Known as *structural realism* or *defensive realism*, **neorealism** was developed by Kenneth Waltz in his seminal work *Theory of International Politics* (1979). Among his key contributions, Waltz theoretically introduced the notion of a *structure* within the international system. According to the author, “any approach or theory, if it is rightly termed ‘systemic’, must show how the system’s level, or structure, is distinct from the level of interacting units. [...] Only by doing so can one distinguish changes of structure from changes that take place within it.” (WALTZ, 1979, p. 40). This structural conception is the theoretical element that distinguishes this school from its classical predecessor.

Neorealism conceives the political structure based on three assertions: (1) the *ordering principle*, or the way in which units are organized; (2) the *differentiation of units* and the specificity of their functions; and (3) the *distribution of capabilities* among units (WALTZ, 1979, p. 88).<sup>37</sup> Although the distribution of capabilities is a key element of neorealism, Baldwin (2013) notes that Waltz never explicitly defines what “capabilities” are, while simultaneously admitting that states could be ranked, for instance, from most to least powerful according to their capabilities. This allows for the interpretation that “capabilities” are, in essence, *resources of power*.

## Offensive Realism

**Offensive realism**, in turn, emerges from John Mearsheimer’s work *The Tragedy of Great Power Politics* (2001). The central idea behind this approach involves two possible modes of state behavior within the balance of power: a state may act either to *defend* the balance of power or to *undermine* it.<sup>38</sup>

This defense or “sabotage” of the balance of power is directly related to *fear*. According to Mearsheimer (2001), fear is grounded in three logics: first, the absence of a supranational central authority capable of protecting states from descending into armed conflict; second, the existence of military power, that is, the offensive capabilities of states; and third, the persistent uncertainty that dominates the international system, since states can never be fully certain about the intentions of others.

These three factors explain the pervasive fear among states within the anarchic international system. As a consequence, states internalize the idea that the more powerful they are relative to their rivals, the greater their chances of survival. From this, it follows that the best survival strategy is to become the *hegemon* of the international system,

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<sup>37</sup>Waltz originally uses the term “capabilities”. According to the Macmillan Dictionary (2017), the word *capability* has two meanings: first, “the ability to do something”; second, “the number of weapons, soldiers, etc., that a country has for fighting a war”.

<sup>38</sup>A powerful state acts to defend the balance of power when the risk of change favors another state. Conversely, it acts to weaken the balance of power when the potential change benefits itself. (MEARSHEIMER, 2001, p. 3). Mearsheimer also argues that the structure of the international system “forces” states — even those seeking only security — to behave aggressively toward one another.

since no other state could seriously threaten a hegemon due to its overwhelming position of power (MEARSHEIMER, 2001, p. 3).

The three theoretical traditions within the discipline of International Relations mentioned above share certain commonalities as well as key divergences.<sup>39</sup>

According to Baldwin (2013), although the “national elements of power” approach remains present in the literature, it faces certain analytical challenges because it treats power as if it were itself a resource. If power is a resource — and resources are static — then what functions as a resource of power in one situation may not function in another, since context is ever-changing. Furthermore, resources are linked to potential power rather than actual power (BALDWIN, 2013).

Thus, the fundamental characteristic of the “national elements of power” perspective lies in the understanding that power can be measured through a state’s resources. Additionally, it assumes that states are capable of translating power derived from their resources into clearly defined national objectives (SHEEHAN, 1996).<sup>40</sup>

## 2.2.2 The Relational Power Approach

The way in which the field of International Relations had observed power until then — through the “national elements of power” approach — changed around the 1950s, when power began to be conceived as the result of a relationship.<sup>41</sup> This is the central concept of the relational power approach. Instead of being grounded in the notion of possession, power now rests on causality (BALDWIN, 2013). This new relational understanding of power is also adopted by Strange (1996), who recognizes the existence of both relational and structural power — as will be discussed later.

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<sup>39</sup>Hans Morgenthau, Kenneth Waltz, and John Mearsheimer all approach power from an essentially military perspective (BALDWIN, 2013, p. 283). Classical realism and offensive realism depict states as actors seeking to maximize power. However, for Morgenthau (1948), this pursuit stems from a desire for power in itself, while for Mearsheimer (2001), the maximization of power is a natural consequence of the anarchic international system in which states exist (BALDWIN, 2013, p. 283). Both defensive and offensive realism portray state objectives as derived from the structure of the international system. Waltz (1979) argues that the primary goal of a state is to maintain a sufficient level of security to ensure its survival, with no need to accumulate power beyond what is necessary. In contrast, Mearsheimer (2001) claims that the ultimate goal of the state is to become a “hegemon,” gathering as much power as possible to reach that status (BALDWIN, 2013, p. 283).

<sup>40</sup>It is worth noting that, within realist perspectives, international politics primarily concerns the politics conducted among the Great Powers — those which Simonds and Emeny (1937) identified as States possessing sufficient military capabilities to effectively sustain the foreign policy determined by their own decisions. Realism does not concern itself with the politics of States that are not considered Great Powers. This is because Great Powers can afford the “luxury” of “waiting to see” how foreign policy unfolds, since their own capabilities — translated into military power — can ensure the implementation and maintenance of such political will.

<sup>41</sup>Baldwin (2013) explains that several scholars viewed the work of Lasswell and Kaplan (1950), *Power and Society*, as a revolutionary milestone in International Relations — precisely for presenting this new conception of power grounded in causality.

The relational power approach assumes that the behavior of actor A, at least partially, provokes a change in the behavior of actor B. It is therefore called “relational,” as it operates through cause and effect within the context of a relationship between two actors. In this view, there is no notion of power derived from resources, capacities, or attributes of the State. Moreover, relational power comprises different dimensions. Baldwin (2013) explains that there are, at minimum, four: *scope*, *domain*, *weight*, and *costs*.<sup>42</sup> According to Jack Nagel (1975), when employing this relational power approach, it is necessary, at the very least, to specify the dimensions of “domain” and “scope.”

There are two major milestones within the relational power approach: the Constructivist school and the “Faces of Power” debate. While Constructivism comprises three relevant studies — the theoretical foundation of the school by Alexander Wendt, the notions of identity and interest developed by Ted Hopf, and the interpretation of “polymorphous” power by Michael Barnett and Raymond Duvall — the Faces of Power debate encompasses three progressive stages, based on the idea that power has three faces. For Baldwin (2013), this was one of the most significant debates involving the concept of power in the twentieth century. Both milestones made important contributions to the understanding of relational power.

## Constructivism

Wendt (1992) identifies the existence of social theories on interests and identities that had been neglected by the discipline of International Relations and incorporates them into a framework he calls “constructivism” — in order to emphasize the focus on the social construction of subjectivity.<sup>43</sup>

Ted Hopf (1998) notes that Constructivism brings alternative themes to the IR agenda: anarchy, balance of power, the relationship between state identity and interest, the elaboration of power, and prospects for change in global politics. Specifically regarding power, Hopf (1998) recalls that Constructivism argues that the notions of *material power* and *discursive power* are both necessary for understanding international affairs.

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<sup>42</sup>“Scope” refers to the aspects of B’s behavior that are affected by A, for example, economic aspects. “Domain” encompasses the set of actors subject to A’s influence. “Weight” refers to the probability that B will be affected by A. “Costs” refer to the means through which B is affected by A, which may be symbolic, economic, military, or diplomatic. (BALDWIN, 2013, p.275).

<sup>43</sup>In 1992, Alexander Wendt published the article *Anarchy is what states make of it*. In this article, he observed that the debate between neorealists and neoliberals shares a common foundation — “rational choice”. This assumption implies that scholars are automatically directed to ask certain questions instead of others, as there is no inquiry into identities and interests. These points are not pondered when discussing actors in International Relations. This means that processes and institutions may change their behavior, but the studies about them preserve their identities and interests — they are static, unquestioned. (WENDT, 1992). Wendt (1992) proposes two avenues of inquiry: first, that the answers to these concerns depend partly on how important the interactions among States are for constituting their identities and interests; second, that they also depend on how easily States’ interests and identities change in response to systemic interactions (Ibid., p.423).

Indeed, States coexist within the international community and maintain relations of power, directly or indirectly. For Constructivism, it makes little sense to focus solely on the material, military, or coercive aspects when seeking to understand interactions among States and the formation of power balances. It is necessary to study the dimension of power that emerges from the relationships among them — relationships that are consolidated through conventionalized practices within the international system (while, for instance, Wendt seeks to understand why certain practices became conventions).

Despite their relevance, States are not the only actors in this system. This curious point reveals that some approaches within the IR discipline aim to bring other actors — beyond the State — into the scope of significant studies and research. Up until then, scholars were primarily concerned with the State and the power games of the international political arena.

The Americans Michael Barnett and Raymond Duvall (2005), in turn, interpret power as “polymorphous” — that is, it manifests itself in different forms.<sup>44</sup> They offer an explanation for why power has been treated, in IR, essentially under the theoretical umbrella of Realism (something that Strange also does).<sup>45</sup> For these authors (Ibid., p.45–46), power can be expressed in two forms: through interaction (that is, through cause and effect); or through the constitution of actors as social beings (in this case, power operates through social relations). They conclude that this conceptual distinction is very similar to the distinction about power in the literature of the field that discusses “power over” and “power to.”

## The Faces of Power Debate

Finally, the Faces of Power debate encompasses considerations regarding the first face, second face, and third face of power. The debate concerning the “first face of power” begins with the studies of Robert A. Dahl (1961) in his work *Who Governs? Democracy and Power in an American City*. In this study, Dahl (1961) seeks to understand who governs in the city of New Haven, in the state of Connecticut, in the United States. In short, the “first face of power” has one focus: to determine who has the power to put forth political proposals in the decision-making process of a given issue. The second face

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<sup>44</sup>The four types of power formulated by them (Ibid., p.43) are: compulsory power, institutional power, structural power, and productive power. Compulsory power is power as relations of direct control interactions of one actor over another. Institutional power is the direct control that actors exercise indirectly over others through diffuse relations of interaction. Structural power is the constitution of the subject’s capabilities in direct structural relation of one with another. And, finally, productive power is the socially diffuse production of subjectivity within systems of meaning and signification.

<sup>45</sup>The authors suggest that realists essentially treat power as a material resource of coercion whose purpose is to influence actor B to alter its behavior. According to them, the tendency to associate power with the Realist theoretical current in IR is due to the fact that realists are practically the only ones who address power directly. Without the existence of considerations about power in other spheres, Realism triumphs almost hegemonically when it comes to the Theories of International Relations. (BARNETT; DUVALL, 2005).

of power is associated with the studies of Bachrach and Baratz (1962): both argue that there are sets of issues that will never be brought to the decision-making table because, in the first face, certain issues were already taken as given. But there are others that are not selected. This neglect reflects the debate of the second face of power: the ability that power has to select which issues will be discussed in the decision-making process. Finally, the third face of power was presented by Lukes (1974) as the manner through which actor A causes a change in the behavior of B. There are different forms — A can affect B's needs, thoughts, desires, and even preferences. Lukes (1974) acknowledges the similarity of the third face of power with the studies of Antonio Gramsci on "hegemony" and Joseph Nye on "soft power."

## 2.3 The Break with the IR Theoretical Tradition

Both through the national elements of power approach and through the relational power approach, the unit of analysis addressed by traditional theoretical currents in IR had been almost exclusively the nation-state. For Strange (1996), this was troubling insofar as it distanced theory from the actual practice of everyday life: non-state actors proliferated in the IS with considerable authority over a range of issues.

When Morgenthau (1948) discussed the balance of power and introduced the view that power could be "measured," he was referring to the power of the State in the international arena: power is amenable to measurement based on a State's resources. Waltz (1979), in turn, understood the international arena through a defined political structure — with an ordering principle, differentiation of units and the specificity of their functions, and the distribution of capabilities among units. In this IS, States can be ranked according to the power they "possess," since power resources are the State's own capabilities. Mearsheimer (2001), for his part, revisited the idea of the balance of power, highlighting the privileged position of the State in this context of the IS. States are the agents that seek to defend their balance of power or to undermine it (if doing so favors them in some way).

In none of the theoretical currents within the national elements of power approach is the relevance of actors other than the State acknowledged. After all, since the elements are "national," these currents neither admitted nor concerned themselves with the existence of elements not attributed to the State. Thus, as sovereign, the State — besides being the sole "player" — also saw in anarchy the reason to secure itself as an independent entity capable of surviving in the IS. These currents did not explain, nor did they intend to explain, the development, the power, or the survival of transnational corporations or any other non-state actor with authority on the international stage. And yet, these currents remained the fundamental pillars upon which the discipline of IR was built — a discipline that, it bears recalling, discussed international politics without, until then, considering non-state political actors.

Shortly thereafter came the relational power approach. It made the observation of power more flexible because it acknowledged, for the first time in the field of IR, that power was not derived solely from national elements. Relational power arose from the conception of an idea of causality: it is the exercise of influence that A has over B so as to alter B's behavior in order to achieve outcomes desired by A. This conception of power broke with the earlier notion grounded in national elements. Whereas power had previously been viewed as a resource, an element of possession, the new approach sought to confer a causal meaning to power, derived from a relationship. But, similar to the national elements approach, analyses employing the relational power approach commonly treated international politics as the arena of States. Broadly speaking, both approaches lacked the recognition and explanation of the existence of power and authorities originating from the Market — or rather, from the private sector. It is here that Strange's contributions become evident. She recognized that, in practice, numerous non-state authorities held power over activities and people in the IS. On this point, she stated that "It seems to me that the powers of most states have declined still further, so that their authority over the people and their activities inside their territorial boundaries has weakened. Non-state authorities, meanwhile, have impinged more and more on those people and their activities." (STRANGE, 1996, p.xi).

How could Strange explain the existence, and the power, of non-state authorities in the IS based on a traditional IR literature? How could she discuss international politics without considering the dimension of non-state actors, recognizing them as authorities with power over activities and people on the international stage? By all indications, she could not. And for this reason, she broke with the discipline of IR to dedicate herself to the study of international politics from the field of IPE. Although there were some approaches that sought to address non-state actors, the fact remained that within the discipline of IR, the State persisted as the center of analyses on international politics. On this point, she stated:

That [new realism] merely added new issues to the agenda of inter-state diplomacy and new bit-players to the cast of actors in international politics. It left the state and its concerns still always at the centre of the stage. It is the *always* that I now find unacceptable, and which leads me to feel that perhaps I have at last reached the final parting of the ways from the discipline of international relations. I have been involved with it now, as student, foreign correspondent and teacher over more than half a century. But I can no longer profess a special concern with international politics if that is defined as a study different from other kinds of politics and which takes the state as the unit of analysis, and the international society of states as the main problematic. (STRANGE, 1996, p.xv)



For Strange (1988), analyses of international politics needed to encompass the practices experienced in the international arena — and this included the existence of non-state actors and the power that these authorities exercised over people and activities. For this reason, she sought to ground her analyses on three fundamental premises:

[...] Politics is a common activity; it is not confined to politicians and their officials. The second is that power over outcomes is exercised impersonally by markets and, often unintentionally, by those who buy and sell and deal in markets. The third is that authority in society and over economic transactions is legitimately exercised by agents other than states, and has come to be freely acknowledged by those who are subject to it. (STRANGE, 1996, p.12–13).

The first premise opens space for political considerations beyond the stage governed by national governments and international organizations — which includes the politics conducted by multinational enterprises, transnational corporations and organizations, and other authorities. The second premise indicates that power does not stem solely from political relations but also from economic and financial relations. Capital opens the opportunity for the exercise of power. The third premise removes the exclusivity of the State's authority over people and activities within its geographical borders. This last premise aims to dismantle, as far as possible, the wide gap between everyday practice and the theories and analyses produced in academic centers around the world. With these three premises in view, the reader can appreciate both the eclecticism of Susan Strange and the relevance of the “symbiosis” between the disciplines of Politics and Economics for the analyses of International Relations — after all, power and capital walk together on the global stage.

In what follows, we turn to the studies on “power” and “diffusion of power” that emerged in IPE from the work of the British scholar Susan Strange, in the late 1980s and mid-1990s. The reviews of the element of “power” discussed thus far help us understand the reasons why Strange (1988) formulated the notion of structural power and how it fits into the reading of power within International Relations. Strange's triumph (1988) was perhaps her contribution to the notion that power also resides in non-state actors within the context of the IS. To this end, Strange (1996) did not resort to the creation of an alternative theory of power that could account for new actors. According to Palan (1999), she employed her own eclecticism to create a framework from which analyses of the practical reality of the IS (and of the foundations that underpin the human condition) could be examined. This framework was conceived by her on the basis of fundamental human values as forms of social organization: wealth, security, freedom, and justice (STRANGE, 1988). In this way, Strange's approach to the element of power proceeds from a holistic perspective — which is why Palan (1999) believes that the concept of structural power cannot be used independently of the framework of the four structures

elaborated by her. After all, structural power does not emanate from the four structures (security, production, finance, and knowledge), but is rather the very articulation of elements originating from them (PALAN, 1999).

In the following sections, we have three objectives: (1) to define the four primary structures described by Strange (1988), with particular attention to the structure of knowledge; (2) to present the concept of structural power; (3) to describe the phenomenon of the diffusion of power. These subjects are found in two main works by the author: *States and Markets*, 1988, and *The Retreat of the State: The Diffusion of Power in the World Economy*, 1996. The latter work is considered by her as “an extension, or elaboration, of the same ideas about power and transnational relations that characterise the contemporary world scene.” (STRANGE, 1996, p.x). The ideas on power and transnational relations are presented in the first work, *States and Markets*.

## 2.4 Structural Power

In 1988, Susan Strange introduced the concept of structural power in her work *States and Markets*. This concept emerges as an alternative interpretation of power to the existing concepts in the IR literature. For her, two forms of power are exercised within a Political Economy: relational power and structural power. However, even while recognizing the existence and exercise of relational power, she points out that structural power has had greater relevance in the IS.

Strange defines structural power and offers several considerations about it. She stated that:

Structural power [...] is the power to shape and determine the structures of the global political economy within which other states, their political institutions, their economic enterprises and (not least) their scientists and other professional people have to operate. [...] In short, [it] confers the power to decide how things shall be done, the power to shape frameworks within which states relate to each other, relate to people, or relate to corporate enterprises. [...] Structural power is to be found not in a single structure but in four separate distinguishable but related structures. [...] These four, interacting structures are not peculiar to the world system, or the global political economy, as you may prefer to call it. The sources of superior structural power are the same in very small human groups, like a family or a remote village community, as they are in the world at large. The four sources, corresponding to the four sides of the transparent pyramid, are: control over security; control over production; control over credit; and control over knowledge, beliefs and ideas. Thus, structural power lies with those in a position to exercise

control over (i.e. to threaten or to preserve) people's security, especially from violence. It lies also with those able to decide and control the manner or mode of production of goods and services for survival. Thirdly, it lies at least in all advanced economies, whether state-capitalist, private-capitalist or a mix of both — with those able to control the supply and distribution of credit. [...] Fourthly and lastly, structural power can also be exercised by those who possess knowledge, who can wholly or partially limit or decide the terms of access to it. This structural power in particular does not easily fit into the layer-cake, club-sandwich model because it may very easily lie in part beyond the range and scope of the state or any other 'political' authority. Yet its importance in political economy, though not easy to define or describe, is not to be underrated. (STRANGE, 1988, p.24–28).

Structural power originates from the four structures mentioned in the previous section. The structures do not emanate power; rather, it is through the articulation of elements arising from these structures that structural power is conceived. For this reason, both Palan (1999) and Brown (1999) do not consider Strange a theorist: she does not create an alternative theory; she provides inputs sufficiently capable of offering a worldview about actors (state or non-state) and their activities based on interrelated structures (themselves conceived from fundamental values of human organization). It is from this structural vision of the IS that actors relate to one another. This structural vision allows, above all, the inclusion of non-state actors in academic studies in IPE — something inconsistent with the discipline of IR up to that point. The relevance of these actors in Strange's work is considerable because (1) it reflects the political and economic practice found in the everyday reality of international affairs, where multinational enterprises (and other organizations or even individuals) maintain agreements and exert influence over governments, people, and organizations; (2) it demonstrates that, within each structure (security, production, finance, and knowledge), the State is neither the sole actor nor the most relevant one to the point of assuming the top position, the leadership, of the structure. On the contrary. In a later work, Strange (1996) acknowledges that the State assumes a leadership position in only one structure: security (STRANGE, 1996). In the remaining structures, the contemporary State is merely one of the relevant actors that exert influence and seek power — it is not the only one.

The four structures are not present solely in the international context. The conception of structural power is grounded in an analytical framework, or rather, a method through which diagnoses are made about the human condition as affected by social, political, and economic circumstances (STRANGE, 1988). Given that such circumstances also describe the local context, structural power is disseminated throughout the economic-political fabric at various levels. Indeed, for this reason, one can analyze, through the

conception of structural power, the human relations that occur in families, neighborhoods, or human tribes — provided that these relations arise from social, political, or economic circumstances.

Furthermore, we observe, from the quotation above, that structural power **falls upon** those who are in a position of **control** within the context of at least one primary structure (security, production, finance, and knowledge). On this point, Strange (1988) stated that:

Structural power lies with those in position to exercise control over (i.e. to threaten or to preserve) people's security, especially from violence. It also lies with those able to decide and control the manner or mode of production of goods and services for survival. Thirdly, it lies — at least in all advanced economies, whether state-capitalist, private capitalist or a mix of both — with those able to control the supply and distribution of credit. [...] Fourthly, and lastly, structural power can also be exercised by those who possess knowledge, who can wholly or partially limit or decide the terms of access to it. This structural power in particular does not easily fit into the layer-cake, club-sandwich model because it may very easily lie in part beyond the range and scope of the state or any other 'political' authority. Yet its importance in political economy, though not easy to define or describe, is not to be underrated. (STRANGE, 1988, p.26–28).

In other words, it is about having control over an object within at least one primary structure — in the local and/or global context. Whether on a neighborhood street, over a neighborhood, or nationally or globally. Moreover, Strange (1988) indicates that the possessor of control over a structure (at its macro or micro level) can exercise power without apparently placing direct pressure on others to make a decision or choose among alternatives. And this is possible because structural power, despite being less “visible” (than material power, for example), is still present.

However, many may question: if structural power derives from articulations originating from four interrelated structures — and not exclusively in an individual manner — how can it fall upon an actor that exercises control over an aspect of at least one structure? Strange does not provide an answer, but we may venture a conjecture. Structural power was represented, metaphorically, by Strange as a tetrahedral pyramid whose faces represent the primary structures (security, production, finance, and knowledge). And, despite originating from the dynamics of elements arising from the structures, it can be “observed” from one of the faces of the tetrahedron, or rather, from one structure. In other words, although it is understood through the four structures (which complement and relate to one another), there exists the possibility of observing structural power originating specifically in the context of one of the structures. It does not originate in isolation

from a single structure, but the analytical focus of structural power can be directed toward one of the structures. It is possible to rotate the pyramid so as to examine each of its faces in detail.<sup>46</sup>

## 2.5 The Four Primary Structures

Susan Strange (1988) defined four primary structures: the security structure; the finance structure; the production structure; and the knowledge structure.<sup>47</sup>

The security structure is the “framework of power” created through the provision of security by some human beings for other human beings (Ibid., p.45). What follows is an example that we have taken the liberty of formulating based on the invitation to creative stimulation made by the author herself in the context of her works.

Imagine a prehistoric era in which human beings wandered in bands across the earth and needed to protect themselves from threats posed by the climate, animals, diseases, geographical hazards, other human groups, and so forth. The individual within the group who was capable of providing security for the other(s) — who, for instance, were being threatened by an enemy — gains “structural power” within this “framework of power.” This individual is capable of affecting outcomes (by protecting the band from the enemy, the outcome is that the group is saved — as opposed to another possible outcome, which is the death of the band) in such a way that their preference (the salvation of the band) takes precedence over the preference of the other (the death of the band). Control over the group’s security confers upon this “savior” individual structural power, which originates through the security structure. Their band, and the enemy band, will come to see this individual as a “powerful” agent and will think twice: before launching violent incursions against them (security); when producing some good that may be of interest to them (production), since there is a risk to the physical integrity of the one who produces the good (after all, to obtain the good, the powerful individual may threaten them); and when providing “loans” or credits in the form of food, clothing, and so on, due to the risks to physical integrity. The relationship between this actor and their own band, and between this actor and rival bands, is altered — even if the “savior” individual is not aware of it.<sup>48</sup> And this relationship is altered because, once they are seen as a “powerful”

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<sup>46</sup>See “Appendix 1 — Pyramid Representing Structural Power.”

<sup>47</sup>There are also secondary structures: transnational transport systems, the trade system, energy supply systems, and the transnational welfare and development system (Ibid., p.139). The British scholar acknowledges that the selection of these structures is somewhat “arbitrary in the sense that it would be equally logical to include some other secondary structures.” (Ibid., p.139). The main characteristic of these secondary structures is that, although they are frameworks of action within which choices are made based on value preferences, they are also secondary to the four primary structures of security, production, finance, and knowledge. When it comes to structural power, she formulates the analogy of the pyramid.

<sup>48</sup>Note here the contribution of gender studies to IPE, as mentioned by Strange (1988).

agent, their mere presence will be capable of operationalizing this power.

At the micro level, this can happen both in the context of prehistory, in small human groups, and in the context of the outskirts of a city in the current era. At the macro level, this occurs both from the domestic standpoint, when the State gathers unto itself the legitimate use of force in society, and from the international standpoint, when it seeks to alter the balance of structural power among actors. According to Strange (1988), in the context of the current international system, the security structure is conceived around the institution of the State, even though the State does not act in isolation within the system (STRANGE, 1988).

The production structure can be defined “as the sum of all the arrangements determining what is produced, by whom and on what terms.” (STRANGE, 1988, p.64). That is to say, from the perspective of Strange’s IPE, the production structure is responsible for creating “wealth.” When this structure changes, “big changes are apt to follow in the distribution of social and political power, and sometimes the nature of the state and the use of authority over the market.” (Ibid., p.64). It is natural that such changes occur, since the production structure lies at the heart of the organization of life in society. Increasingly, the production structure in the global context is characterized by transnational enterprises. On this point, Strange (1988) stated that “the nature of the global production structure [...] is the combined result of state policies and of market trends, of management strategies and changing technology.” (Ibid., p.80).

The finance structure is defined according to the actors capable of creating or denying the granting of credit to others, without whom no advanced economy can function (STRANGE, 1988). If we observe this issue from the micro level, we perceive that “the power to create credit implies the power to allow or to deny other people the possibility of spending today and paying back tomorrow,” and at the macro level, “all the arrangements governing the availability of credit plus all the factors determining the terms on which currencies are exchanged for one another.” (STRANGE, 1988, p.90). Reflexively, the one who has control over credit is capable of directly or indirectly affecting the scope of the choices available to others.

The knowledge structure is defined by Strange (1988) as follows:

[...] A knowledge structure determines what knowledge is discovered, how it is stored, and who communicates it by what means to whom and on what terms. [...] So power and authority are conferred on those occupying key decision-making positions in the knowledge structure — on those who are acknowledged by society to be possessed of the ‘right’, desirable knowledge and engaged in the acquisition of more of it, and on those entrusted with its storage, and on those controlling in any way the channels by which knowledge, or information, is communicated. (STRANGE, 1988, p.121).

This structure is related to knowledge that is produced, discovered, stored, and communicated. Note that Strange (1988) does not draw conceptual distinctions between “knowledge” and “information,” so long as the unit in question is considered socially relevant. Those who possess relevant and desired knowledge, and also those who in some way control the channels of communication and storage of such information, are conferred structural power originating from the knowledge structure. Beyond power, they are also conferred authority.<sup>49</sup> These two elements, power and authority, are conferred upon these individuals by virtue of occupying important positions within the structure, because in addition to being decision-makers, they are also recognized as holders of the knowledge desired by society (which, therefore, has value for the social organism). We may assume some examples originating from the knowledge structure, such as researchers, content creators, library science professionals, information systems specialists, computer scientists, and so on — in short, the class of professionals (and prominent figures in a given area of knowledge) who are related, directly or indirectly, to the production, communication, and storage of knowledge (or information) considered socially relevant. These figures are considered “authorities” within the knowledge structure. And to be an authority, it is necessary, first, that the actor actually possesses the said relevant knowledge; and second, that such knowledge be considered important (STRANGE, 1988).

Some examples presented by Strange (1988) highlight past actors who succeeded in the accumulation of structural power, originating mainly from the knowledge structure, over the years: the Catholic Church during the medieval period and the scientist State of the twentieth century. The Catholic Church held authority and power because, among other important points, it controlled the means of communication (which spanned all of medieval Europe) through which knowledge was transmitted, in the form of sacred books and literacy in the sacred Latin language. Those who were granted access to the storage of knowledge — the sacred books guarded by the Church — needed, in addition to access, knowledge of the language in which the information was inscribed. Note that it was the Catholic Church that controlled, during the medieval era, the means of communication and storage — in addition to investigating discoveries and denying access to knowledge to other individuals. The scientist State, on the other hand, was identified by Strange (1996) as the institution that assumed some of the Catholic Church’s responsibilities regarding the knowledge structure (such as the growth of the educational system that allowed access to education in schools and universities) and invested in technical innovations (through which new knowledge was discovered in various fields, especially Physics), thereby giving rise to intellectual property rights (STRANGE, 1988).

The knowledge structure is also somewhat different from the other structures for

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<sup>49</sup>This semantic distinction between “power” and “authority” in this quotation from Strange is relevant for understanding authority as a dimension in its own right. This will be discussed at a later point.

several reasons: (1) Strange (1988) affirms that the power originating from the knowledge structure is understandably diffuse; (2) it is a power she describes as “unquantifiable”<sup>50</sup>; (3) the power derived from the knowledge structure is also grounded in the denial of access to knowledge; (4) it is a power more associated with consent than with coercion — unlike the other structures; (5) there must exist social recognition that a given piece of information is considered important — so that the person holding this knowledge is also considered relevant to society (STRANGE, 1988).

Finally, we must understand that the structures are not static; they are dynamic and subject to changes and developments. On this point, Strange (1988) acknowledges that, of all the structures, the knowledge structure is the one undergoing the most rapid changes at the end of the twentieth century.<sup>51</sup> And although she did not write specifically about the Internet when making considerations about the knowledge structure, Strange recognized the relevance of the information revolution occurring at the end of the twentieth century.<sup>52</sup> In particular, she highlighted the importance of the combination of three technological areas that were undergoing major changes: widely available and low-cost computer systems; large-scale satellite communication systems in orbit; and the digitization of language, which opens the potential for the dismantling of linguistic barriers among human groups (STRANGE, 1988). And, in the end, she concluded that the interaction among the structures suggests very important political conclusions: the ongoing

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<sup>50</sup>My interpretation of this is that in the other structures, the power originating from them is still amenable to quantification — even if roughly. In the finance structure, for example, it is possible to have a sense of the “quantity” of power through a banking analysis of credit granting. From a micro perspective, one can gain a sense of structural power by examining how much was “lent” to an individual. This financial amount determines the purchasing power of the individual who received the credit and offers an idea of the intensity of the creditor’s power. In the production structure, by contrast, we can have a reasonable idea of the “quantity” of power through how much was produced (or left unproduced), how much was manufactured. In the security structure, this perception of “measurable” power occurs through the analysis, for instance, of the size of armies, the quantity of a State’s military resources, the number of nuclear warheads. The possession of 50 nuclear warheads versus “none” is an indicator, broadly speaking, of the power that certain States hold in the provision of security or the threat of violence. At the micro level, we observe the number of members comprising a street gang and the quantity of resources they possess both to protect and to threaten third parties. But in the knowledge structure, it is understandably difficult to quantify “how much” knowledge an individual, or group of individuals, possesses. Knowledge and information do not translate into numbers of books, articles, journals, newspapers, and so on, since these numbers do not automatically signify absorbed and communicated knowledge. This is the difficulty of measuring structural power originating from the knowledge structure. We should note, however, that Strange (1988) indicates that control over the channels of communication (and also storage) through which information, knowledge, and beliefs are communicated also confers structural power. Thus, the one who allows or denies others access to these channels of communication holds power. This point is perhaps the only one that permits an illustration of the “quantity” of power originating from the knowledge structure: a global private company that, through the use of technology, enables millions of people to communicate using its communication channels (e-mails, servers, chat platforms, shared “office” suites, etc.) and decides to alter its access policy holds structural power, as it directly affects the options of action available to its clients. Even so, it is merely an indicator of power.

<sup>51</sup>Strange writes about the primary structures in the context of her 1988 work, that is, also at the end of the twentieth century.

<sup>52</sup>Susan Strange passed away in 1998, and thus did not write specifically about the digital information systems that culminated in the rise of the commercial Internet.



technological innovations were resulting in the global unification of goods and services (Ibid., p.130). Although the world had already experienced the unification of markets, these were limited by transportation systems. The new dynamic, grounded in microelectronics, is different: the new means of communication allow information to be accessible to buyers and sellers globally — and they also allow purchase and sale decisions, as well as the execution of those decisions, to be carried out instantaneously. Therefore, this entails innovative political and economic consequences (Ibid., p.131).<sup>53</sup>

Strange (1988) demonstrates awareness of her limitations regarding the knowledge structure. In part because she recognizes that the power derived from the knowledge structure is the most neglected of all (security, production, and finance) in the studies of social psychologists, philosophers, technology specialists, and others. As such, there is little input for analyses that depart from the knowledge structure. And in part because the change occurring in the knowledge structure is the fastest among all structures and, for this reason, the results are not yet clear. But, despite this, she asserts that there is sufficient evidence to sustain that the knowledge structure is undergoing at least three broad transformations.

The first of these has to do with the very relevance of the knowledge structure. Strange (1988) affirms that at the end of the twentieth century, the knowledge structure is considered so important that competition among States in the IS can be understood as a competition for the leadership of this structure. In the past, this was not the case: States competed for territory. And the competition for territory occurred because the production of wealth (and, therefore, the acquisition of power) depended on land and natural resources. The twentieth century altered this competitive conception: States seek the leadership of cutting-edge technological development (Ibid., p.136). According to her, there has long been popular recognition that “knowledge is power,” but IR and IPE theorists have yet to absorb this reality within academia.

The second development refers to the increase in asymmetry among States as political authorities in the acquisition of knowledge and access to it (Ibid., p.137). In the IS, States present themselves as sovereign and equal political authorities — but access to and the acquisition of knowledge do not occur on equal terms.

The third development, less explored by the author, concerns new distributions of power, social status, and influence beyond state borders — by reason of the knowledge

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<sup>53</sup>Some of these economic and political implications include: the relevance of information for the production structure is significantly increased, and with it, the information revolution devalues the power and wealth of industrial workers (that is, fewer people would work on the factory floor, in mines, on farms, on ships, and so on, and more people would be placed in offices with computers and processors — fewer “blue collars” and more “white collars”); large manufacturing companies would be compelled to diversify into information sectors; within companies, the value of the “knowledge worker” is increased at the expense of the value of industrial workers; the power and capabilities of management in large companies are enhanced; companies would be able to significantly increase the internationalization of information; and so on (STRANGE, 1988, p.131–133).

structure (STRANGE, 1988). According to her, for example, it is the number of nationals with access to higher education that differentiates States. The accumulation of education, information, and knowledge removes obstacles to access to credit (and not merely the accumulation of wealth in any form). And she further affirms that a shift in the transmission of power is taking place: gradually, power is moving from nations “rich in capital” to those “rich in information.” For this reason, she considers that access to knowledge lies at the heart of social status and the distribution of power (Ibid., p.138).

## 2.6 Diffusion of Power

Having discussed relational power, structural power, and the primary structures defined by Strange (1988), it is now appropriate to present the phenomenon of the diffusion of power. This phenomenon is the subject of the work *The Retreat of the State: The Diffusion of Power in the World Economy*, from 1996. Although the expression “diffusion of power” appears in the title, it appears only once throughout the entire content of the work.<sup>54</sup> However, the term “diffusion” quite commonly appears associated with the word “authority.”

According to Strange (1996), it is difficult to define authority. Explicitly, she does not define this concept in either of her two works. However, there are two distinct passages that elucidate its meaning. First, (1) in *States and Markets*, when discussing the broader use of the term “politics”: “[...] Extending the definition of politics beyond states to all sources of authority, to all with power to allocate values, however, allows the two worlds of markets and states, of government and business, to be treated as one, rather than as two as in Gilpin’s equation.” (STRANGE, 1988, p.38, emphasis ours). And second, (2) in *The Retreat of the State*, when investigating how to identify who holds authority in the global political-economic arena: “[...] The first, basic question was ‘Who, or what, is responsible for change?’ The second was ‘Who, or what, exercises authority — the power to alter outcomes and redefine options for others — in the world economy or world society?’” (STRANGE, 1996, p.184, emphasis ours).

In the first passage, authorities are those entities with sufficient power to allocate values in the social, political, and economic context. These values, according to Strange (1988), are those arising from human social organization: security, wealth, justice, and freedom of choice — considered by her as the four basic values of political economy (Ibid.,

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<sup>54</sup>She uses the expression “diffusion of power” when discussing telecommunications companies. We highlight the following excerpt: “Yet in most societies, the political contest has always been between those who gave priority to the short-term advantages of financial and technological bargaining power and those who saw the longer-term advantages of social and political legitimacy as a result of their enlightened concern for the broader interests of civil society. In our own times, that contest continues — but this time on a global scale. Here, the diffusion of power among so many governments, and from them to non-state authorities makes it more difficult for policy-makers to take the long, more socially and economically enlightened view.” (STRANGE, 1996, p.108–109).

p.5).

In the second passage, we can identify what authority means for Strange (1996). She makes clear that the authorities to which she refers are entities that have sufficient power to alter outcomes and define options for others — which is consistent with the first passage, but adds the relational aspect of influence over outcomes. By definition, therefore, within the analytical framework developed by IPE, the existence of authorities other than the State is presumed: the State is not the only entity with the power to allocate values, redefine options for others, or influence outcomes. The great test, however — that which determines whether an entity has authority (in fact, and not merely in potential) — is the outcomes (STRANGE, 1996, p.91). In other words, perceiving a non-state actor as an authority, within a given context, is an exercise in analyzing its influence over outcomes.

According to Strange (1996), in the political-economic landscape of the twentieth century, the power to allocate values and alter outcomes for others was essentially concentrated in the figure of the nation-state. The beginning of the concentration of authority in the nation-state started with the Treaty of Westphalia. Before that, a great deal of authority was concentrated in the Catholic Church.<sup>55</sup>

The diffusion of power is the inverse phenomenon: the power of the nation-state — both in allocating values and in exercising power over outcomes, from the analytical framework of IPE — diffuses throughout the social, political, and economic fabric to other authorities that at times rival the State and at times reinforce it.<sup>56</sup> In other words, “[...] The declining authority of states is reflected in a growing diffusion of authority to other institutions and associations, and to local and regional bodies, and in a growing asymmetry between the larger states with structural power and weaker ones without it.” (Ibid., p.4). And, especially at the end of the twentieth century, the phenomenon of the diffusion of power emerged prominently: “[...] Authority over society and economy is

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<sup>55</sup>According to her, “The Treaty of Westphalia of 1648 is familiar to all students of international relations as the benchmark of a new era in which the authority and sovereignty of the state would be unchallenged. By implication, it marked the virtual end of the constraints imposed on kings and princes by the Church.” (STRANGE, 1996, p.125).

<sup>56</sup>The classification of these authorities occurs only in relation to the state actor itself. Strange (1996) suggests classifying existing authorities in relation to the authority of the State itself. In this way, let us imagine a continuous line on which, at one extreme, is placed the non-state authority that contests and challenges (or even threatens to replace) the authority of the State. And at the other extreme of this same line, the non-state authority that sustains and reinforces the authority of the State. In the middle of this line are placed authorities that exercise either reinforcement or rivalry in relation to state authority. Whether the authorities lean toward one end or the other of the line is determined by the State, since the perception of rivalry or reinforcement is made by the State itself. This perception is quite subjective and subject to change (STRANGE, 1996, p.92). From this, we suggest some examples: nightclub bouncers, professional street security guards, and private security companies are examples of authorities (formal yet non-supreme) within the security structure that sustain and reinforce the authority of the State by providing protection services to other actors (which may be individuals, groups of people, companies, properties, etc.). An example of rivalry, in this case, would be the so-called “parallel state,” that is, organized criminal groups that offer protection and threaten violence against individuals, groups of people, companies, properties, and so on.

undergoing another period of diffusion after two or three centuries in which authority became increasingly centralised in the institution of the state.” (Ibid., p.86–87).

Therefore, if we observe each of the primary structures that compose the analytical framework of IPE, we perceive that authority is shared with other actors. In sum, we can also identify the phenomenon of the diffusion of power, in synthesized form, in the hypothesis of the 1996 work, in which she states:

My hypothesis, as explained in the last chapter, is that on many issues most states have lost control over some of the functions of authority and are either sharing them with other states or with other (non-state) authorities. The outcome in some cases is that no one is responsible for authority functions, even though they may pretend to be. It presumes some general decline in the power of most states and some gain in the authority of world markets and of enterprises operating in world markets. This shift away from states and towards markets is probably the biggest change in the international political economy to take place in the last half of the twentieth century. [...] I shall argue that one of the major shifts resulting from structural change has been the increased power and influence of the multinationals — more properly called transnational corporations (TNCs) — and the networks they set up and operate. (STRANGE, 1996, p.42–43)

The phenomenon of the diffusion of power presented by Strange (1996), according to her, explains political-economic movements that occurred in the second half of the twentieth century, such as the global scenario following the oil shocks and inflation. Traditional IR currents did not seek to explain this moment. In particular, the focus of her narrative concerns the U.S. economy. According to the British scholar, during this period, the American economy both grew and “spread” throughout the world, as did the sources of its power — which shifted from “land” and “people” to settle upon control over the structures of the global system.

In detail, Strange (1996) discusses the disorder caused by the inflations that followed the oil shocks from a global structural perspective — which in this case is the IS in which both state and non-state actors operate. There is recognition of the relevance of non-state actors, since they are imbued with structural power. However, the discipline of IR, and in particular the traditional theoretical currents within the field, cannot explain this same scenario because they commonly attribute to the State the unit of political analysis. It is this political perspective — that the State is the supreme actor in international politics — that collides with the economic perspective, since the latter acknowledges the relevance of non-state actors as subjects of analysis. For this reason, any IPE analysis must expand the unit of political analysis, for if the locus of power migrates from “land” and “people” to “control” over elements originating from the structures of the global system, the actor

in possession of this control is said to be an “authority” within this structure.<sup>57</sup> On the importance of admitting analyses of non-state actors, Strange (1996) argues that:

Extending the focus of analysis from states to all forms of authority allows us to ask how, and by whom values are allocated and political decisions taken — in the wider sense outlined above — to affect outcomes. At one and the same time, we can ask about authority within states and outside them as well as just in their relations with each other. We can avoid the perennial temptation in the study of international relations to ‘reify’ the state, that is, to treat it as one ‘thing’, a unitary actor, as if France, say, or Japan, were a discrete personality. (STRANGE, 1996, p.37).

By extending the focus of analysis from the State to other forms of authority, IPE separates itself from the discipline of IR in its traditional sense. The adoption of a more encompassing definition of politics, which takes into consideration all forms of authority (including all those capable of allocating values), allows the union of Politics and Economics so that they may be treated as a single whole rather than as two separate areas.

## 2.7 The Three Dimensions of the Diffusion of Power

When Strange (1988, 1996) discussed structural power, primary structures, authorities, and the diffusion of power, she left an open invitation for others to explore the subject through reading, informed discussion, and disciplined thinking. In her 1988 work, she affirms that the book is not a conventional “textbook” because it would suggest ways of thinking about politics within the scope of the world economy. For this reason, we cannot call her interpretations a theory, but rather an analytical framework. On this point, she stated that:

This is not a conventional textbook. Students are often given books to read which tell them what they are supposed to know, or else what they are supposed to think. This is not like that. It is going to suggest to you a way to think about the politics of the world economy, leaving it to you to choose what to think. [...] Before you there is not a set menu, not even an a la carte menu, but the ingredients for you to make your own choice of dish and recipe. This is partly because I believe profoundly that the function of higher education is to open minds, not to close them. The best teachers are not those who create

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<sup>57</sup>But not necessarily the supreme authority: it may be the supreme authority (for example, Strange affirms that the State is the supreme authority in the security structure) or occupy a position below the supreme one (a mafia boss is an authority in the neighborhood where they reside when providing security to residents in exchange for their silence).

in their own image a crowd of uncritical acolytes and followers, obediently parroting whatever they say or write. The best are those who stimulate and help people with less experience in and exposure to a subject than themselves to develop their own ideas and to work them out by means of wider reading, more informed discussion and more disciplined thinking. (STRANGE, 1988, p.9).

As a reader and student of Strange, her works stimulated me to think about the possibility of operationalizing the concept of diffusion of power. Moreover, I felt that there would be no inconsistency in seeking, within her literature, clues that could assist me in deepening my understanding of IPE and in attempting to operationalize the phenomenon — given that she herself, as a teacher and author, extends this invitation and encourages these activities. For this reason, after analyzing her text and reflecting on the conceptions she presented, I believe that the diffusion of power operates across three dimensions — which I have called “authority,” “control,” and “outcomes.”<sup>58</sup> These three dimensions are grounded in Strange’s own literature (1988, 1996), so that, in the process of operationalizing the phenomenon of diffusion, these three dimensions could be understood as “discrete qualitative variables” and, therefore, capable of receiving numerical values.

The next section of the chapter is organized as follows: (1) we seek to substantiate, from the literature, the reasons that lead us to believe that the diffusion of power operates across three dimensions — with emphasis on excerpts from the works that allow for this reflection; (2) to present the “valuations” of each dimension when translated into “variables” — that is, what values each of the three variables can receive and why.

### 2.7.1 The “Authority” Dimension

When the reader undertakes a careful reading of the two works by Strange mentioned in this dissertation, they have the capacity to observe that three elements are found very close to the core of the phenomenon of the diffusion of power — which, as is known, opened the way for analyses of the IS from the field of IPE. These three elements are authority, control, and outcomes.

*Authority* is a concept whose definition is quite complicated, as acknowledged by the British scholar herself. However, we can affirm that, in *The Retreat of the State*, in virtually all statements about the “diffusion of power,” Strange used the term “diffusion of authority” — as if they were synonymous and interchangeable terms. Our interpretation is that they are not.

The term “authority” has, in our view, two interpretations that, despite being quite subtle, are sufficient to distinguish them: the use of the term “authority” sometimes

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<sup>58</sup>These words are the respective translations of “authority,” “control,” and “outcomes” — words used by Strange in the context of her two works.

describes an entity that commands the trust of its peers and stakeholders (*being* an authority), and sometimes describes the exercise of power (*exercising* authority). We believe that the expression “diffusion of authority” (which Strange used in her work as nearly synonymous with “diffusion of power”) is linked to the second interpretation, that is, the diffusion of the exercise of power (exercising authority). This is because if it referred to the diffusion of “being” an authority, this would translate into the “dismemberment” of the State, as an authority, into various other entities with lesser authority. In other words, the diffusion of authority, in this case, would mean the emergence of a plurality of lesser authorities, originating from the State and detached from it. And this is not what the phenomenon of the diffusion of power is about.<sup>59</sup>

Being an authority within the IS, viewed through the lens of the four primary structures, demonstrates that the actor has gathered sufficient authority to exercise it in a way that affects outcomes. We assume that this gathering of authority is granted by peers and stakeholders in two ways: via consent or subordination. An authority within the security structure is the State. Its peers (other States) and stakeholders (its nationals) formally grant it authority so that it may be recognized as sovereign in the IS, and thus flow the policies and agreements that derive from the recognition of sovereignty. Another authority within the security structure is the Italian mafia, illustrated by Strange (1996) through the organization “Cosa Nostra.” This organization is recognized by its peers (other mafia groups) as an authority. Citizens who are directly affected by the activities of this organization (for example, inhabitants of a neighborhood with a high concentration of “Cosa Nostra” members) may not consent to the organization’s authority (many do consent), but all subordinate themselves to this authority if they wish to avoid conflict. Both authorities, the “Italian government” and “Cosa Nostra,” clash and conflict on the domestic stage because they rival each other. The Italian State views the mafia organization as a rival to its authority, and conflicts, grounded in the rule of law, ensue. In this example, what distinguishes the mafia organization’s authority from the State’s authority is formality. The Italian government is the authority that concentrates the identity of the Italian State through formal means — recognized through the democratic vote, to which the armed forces, other institutions, and citizens subordinate themselves. Similarly, the Italian national football team during a World Cup also concentrates, in the sporting domain, the identity of the State — but informally. The Italian mafia, on the other hand, is an informal and illegal authority.

Exercising authority is exercising power. This power is exercised directly through relational power and indirectly through structural power (STRANGE, 1996, p.91). In

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<sup>59</sup>We have highlighted the passages in which Strange (1996) writes about “authority” and grouped them into two categories: passages that present authority as the exercise of power; and passages that present authority as an institution, as “being” an authority. These two groups can be found in the list of appendices. We recommend consulting “Appendix 1 — Being an Authority” and “Appendix 2 — Exercising Authority.”

the example given above, both authorities — the Italian State and the mafia through the “Cosa Nostra” organization — exercise power because they influence outcomes for a set of individuals and institutions and redefine their options of choice. The Italian government is the supreme political authority within the territorial limits of the Italian State, which holds the legitimate use of force. The mafia, on the other hand, organizes itself through structures that exert influence over a set of activities and over the security of citizens, as they often act through coercion and violence.<sup>60</sup> While one is a formal authority, the other is an informal authority. But both exercise authority, both exercise power. One of the objectives of scholars in the field of IPE is “to try and untangle the complex web of overlapping, symbiotic or conflicting authority in any sector or on any who-gets-what issue.” (Ibid., p.99).

Being an authority and exercising authority, although very similar, entail distinct yet complementary consequences. An actor can gather sufficient trust from its peers and stakeholders so as to be considered an authority on a given matter. This trust may have been granted through consent or through imposed subordination. In the first case, we can cite as an example an international organization whose members are States. At the outset, a group of States granted sufficient authority, through expressed consent, for the General Secretariat of this organization to begin its activities on a given matter. Over time, other States, observing the functioning of this organization, decided to participate as members. All of these member States are responsible for granting authority to the organization, which develops a series of binding agreements on the matter in which it specializes (in this case, it is recognized as an authority within the knowledge structure), which must be followed by all members.

Another way of gathering the trust of peers and stakeholders is through imposed subordination. Let us imagine what happens in the favelas dominated by drug trafficking in the city of Rio de Janeiro, Brazil. Distinct factions compete for power and shares of the drug market. Let us imagine that faction A holds authority over the hillside community and imposes a set of rules on the citizens who live there — such as identifying oneself every time one enters or leaves the neighborhood and not circulating after 10 p.m. Even though they do not consent to this set of rules imposed by the dominant group on the hill, without alternatives, the residents recognize faction A as an authority (in this case, within the security structure) and abide by its rules so that, in return, they do not face violence. In this scenario, faction B emerges, interested in dominating the hill and reaping profits from the drug trade. A “war” begins between the two groups, the consequence of which is the victory of faction B. The consequence of this is that faction B is now recognized as

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<sup>60</sup>Strange (1996) highlights the suggestion of some sociologists regarding the origin of mafia authorities, stating that “Sociologists have argued that criminal gangs, like underground resistance movements in wartime or recalcitrant groups in prisons, tend to emerge when state authority, for whatever reason is already weakened, and the government has lost or failed to obtain the consent of the governed.” (STRANGE, 1996, p.116).



the new authority on the hill. The residents of the neighborhood subordinate themselves to this new group.

In the first example, authority was recognized by the member States without the organization exercising power in a way that affected outcomes. In this first context, “being an authority” emerges before “exercising authority.” In the second case, things change. Faction B “exercises authority,” or rather, exercises power in a way that affected outcomes — it eliminated faction A and was recognized as the new authority on the hill. “Being” an authority and the exercise of authority are moments that complement each other — regardless of which moment came first. Exercising power reinforces the authority of the actor in question. However, it is important to emphasize that the actor tends to lose authority (“being an authority”) when it is no longer capable of exercising power in a way that affects outcomes. This is at the heart of the issue raised by Strange (1996) when she stated that the authority of a non-state actor could only be determined on the basis of outcomes. It is of no use to be an authority without possessing the capacity to exercise that authority through power — over time, the trust of peers and stakeholders dilutes and the recognition as an authority on a given matter fades away.

Some further considerations are warranted regarding the recognition of authority through the trust generated by peers and stakeholders — specifically within the knowledge structure. Examples of this are expressed in Strange’s literature (1988) in at least three instances. In Hindu society, religious authority was sustained by belief. This was capable of maintaining a set of rules over society regarding diverse activities that ranged from personal hygiene and marriage to the consumption of food through typical diets (STRANGE, 1988). The religious authority, represented by the “brahmins,” therefore gathered the trust of peers and stakeholders such that segments of society voluntarily followed the precepts determined by the religious authority. It should be noted that this authority is located primarily within the knowledge structure.

On another occasion, speaking explicitly about the knowledge structure, Strange (1988) points out that authority and power are conferred upon those who “are acknowledged by society to be possessed of the ‘right’, desirable knowledge and engaged in the acquisition of it, and on those entrusted with its storage, and on those controlling in any way the channels by which knowledge, or information, is communicated.” (Ibid., p.121). The recognition of authority within the knowledge structure — such as, for example, the authority of individuals who are part of the scientific community — is granted by society, which, to a certain extent, trusts that these members possess the knowledge they express. A researcher in the field of IPE, for example, holding a doctoral degree, is recognized by society as having knowledge on certain matters of this research agenda. The degree, conferred by an educational institution and recognized by a government body (representative of society in matters of education), is the granting of authority on the subject of IPE. It is from this granting, from this trust placed in them, that the researcher in this example

may enjoy a set of specific legal rights and prerogatives. It should be noted that, in this case, the authority is formal.

Finally, in a third instance, Strange (1988) comments on the trust of peers and stakeholders in the formation of authority within the scope of the World Administrative Radio Conference (WARC). In this case, she considers the hypothesis that the international bureaucracy responsible for WARC acted according to its own policies — that is, independently of state policy. She affirms that “[...] only if there is real evidence that the agency has some technical capability, or some special authority legitimated not by the approval of governments but by the consent and respect of those affected should it be considered separately from state policies.” (STRANGE, 1988, p.233). Should this hypothesis be confirmed, it can be affirmed that the trust of stakeholders was a determining factor in the legitimation of this international bureaucracy’s authority within the scope of the matters addressed by WARC.

These three examples are important for the discussion of this dissertation because they legitimize that authority, especially within the knowledge structure, is conferred through the trust of peers and stakeholders in the matter over which the authority claims to have knowledge. The perception of the degree of trust of peers and stakeholders — increasing trust, stable trust, decreasing trust — bears upon the “authority” variable present in the database generated for this research.

## 2.7.2 The “Control” Dimension

Like authority, /textitcontrol is also a highly relevant element for understanding the diffusion of power. We consider “control” to be one of the three dimensions of the diffusion of power. In *States and Markets*, Strange (1988) points to the role of control in the formation of power — with special emphasis on structural power, but also pointing to economic and political power, stating that:

Banks, by controlling credit, have economic power. Equally, we can say that people have political power if they control the machinery of state or any other institution and can use it to compel obedience or conformity to their wishes and preferences from others. The trouble with this distinction, however, is that when it comes to particular situations — particularly in the international political economy — it is very difficult (as some later examples will show) to draw a clear distinction between political and economic power. (STRANGE, 1988, p.25)

Here we can perceive that control over an object relevant to the achievement of desired outcomes lies at the core of economic and political power. In the case of banks, the object is credit, for it is control over it that allows banks to have sufficient power to influence

outcomes in the economic arena.<sup>61</sup> Equally, control over the state apparatus by a political group, for example, confers upon it sufficient power to influence outcomes in the political arena — at least to a certain extent.

Similarly, we examine the relevance of control in the context of the four structures from the following quotation by Strange (1988) on the sources of structural power:

These four, interacting structures are not peculiar to the world system, or the global political economy, as you may prefer to call it. The sources of superior structural power are the same in very small human groups, like a family or a remote village community, as they are in the world at large. The four sources, corresponding to the four sides of the transparent pyramid, are: control over security; control over production; control over credit; and control over knowledge, beliefs and ideas. Thus, structural power lies with those in a position to exercise control over (i.e. to threaten or to preserve) people's security, especially from violence. It lies also with those able to decide and control the manner or mode of production of goods and services for survival. Thirdly, it lies — at least in all advanced economies, whether state-capitalist, private-capitalist or a mix of both — with those able to control the supply and distribution of credit. Such control of credit is important because, through it, purchasing power can be acquired without either working for it or trading for it, but it is acquired in the last resort on the basis of reputation on the borrower's side and confidence on the lender's. Fourthly and lastly, structural power can also be exercised by those who possess knowledge, who can wholly or partially limit or decide the terms of access to it. (STRANGE, 1988, p.26, emphasis ours).

In other words, the source of structural power resides in the control of objects originating from the primary structures. In this regard, our attention turns especially to the knowledge structure and the role that control plays in the formulation of power originating from it. Here, it bears recalling once more that the power originating from this structure is also conditioned by control over the channels through which information circulates and is stored (STRANGE, 1988). Let us consider an example of this dynamic.

We can imagine an institution dedicated to research on marine animals. All the knowledge produced through research, data collection, and sample analysis rests upon prior knowledge on a given subject. Upon inferring that the whale is a mammal and not a fish, the researcher can devote themselves to analyses of problems typical of mammals,

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<sup>61</sup> Another example, to illustrate the role of control in power, refers to Great Britain's political control over India. This political control allowed London to extract gold annually and thus use control over the gold supply to influence the Pound-Rupee exchange rate and other instruments of exchange rate policy. British capital flows were essential for the maintenance of economic growth in the pre-1914 world (STRANGE, 1988, p.101).

such as the development of milk in this specific species of whale, for example. But they can only do so if they know that the animal is a mammal and that mammals possess mammary glands. Someone, at some prior moment, produced the knowledge that whales are mammals that inhabit the oceans. This information was stored so that, from it, new studies followed in succession. Furthermore, this knowledge was communicated and shared among various members of the scientific community.

Now let us imagine that, by terrible fate, a great catastrophe befell a large part of human knowledge about animals of all species, and that this institution dedicated to research on marine animals, along with a handful of scientists, was one of the few to be spared from this terrible fate. Let us also imagine that all the knowledge produced and stored by this institution resides in the “cloud” — that is, in some abstract storage location on a computer server. In this hypothetical scenario, the senior researcher of this institution and the IT professional hold great authority and structural power because they control access to the information channels — the researcher by virtue of having experience and knowledge recognized by peers; and the IT professional by providing access to the storage of this knowledge through the channels where information circulates and is stored, that is, in the cloud and in the digital computer network. Both are important for the society that carries on after the catastrophe. Equally, other individuals and entities (such as the hypothetical institution) in this scenario may have the capacity to gather sufficient authority on a given matter due to the knowledge they possess — such as, for example, knowledge of survival in nature. For, as Strange (1988) said, “a knowledge structure determines what knowledge is discovered, how it is stored, and who communicates it by which means to whom and on what terms.” (STRANGE, 1988, p.121). From this, we can infer that those who control objects (1) determinant for the type of knowledge that will be researched/discovered, (2) of the type that store information, (3) that form communication channels through which knowledge is transferred — these are relevant objects within the knowledge structure and confer upon those in possession of their control with power. For, “[...] Knowledge is power and whoever is able to develop or acquire and to deny the access of others to a kind of knowledge respected and sought by others; and whoever can control the channels by which it is communicated to those given access to it, will exercise a very special kind of structural power.” (STRANGE, 1988, p.30)

Some examples of entities that, to some extent, controlled objects considered important within the knowledge structure and ascended to power are highlighted in Strange’s literature (1996): the Catholic Church in medieval Christian Europe; the scientificist State during World War II; companies that hold a monopoly over a specific technology; among other cases.<sup>62</sup> Control over a given object represents power — which, naturally,

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<sup>62</sup>We have compiled in a table examples from both of Strange’s works (1988, 1996) that refer to “control” and the “object” of control. It is possible to verify that in some instances she points out that “control” represents power. See “Appendix 3 — Control, 1988” and “Appendix 4 — Control, 1996.”

influences outcomes. For this reason, control is one of the dimensions of the diffusion of power.

### 2.7.3 The “Outcomes” Dimension

*Outcomes* constitute the third dimension of the diffusion of power. This dimension is intimately related to the dimensions of authority and control because it represents the existence of the effectiveness of both. We may assume that a given actor, in the context of IPE, holds authority over a matter. Through this authority, obedience and subordination are conferred upon them. We may also assume that this actor holds control over an object considered relevant, by peers and stakeholders, in the political-economic arena. Consequently, this actor is considered important within the context of IPE. In both situations, what attests to the actor’s authority, and what attests that it holds control over the object, is its capacity to influence outcomes. That is, if it is verified, through the outcomes, that the actor had the capacity to influence the status quo, its authority is reinforced and/or its control is confirmed. If its capacity to influence outcomes is not verified, its authority is shaken — which means that the trust of peers and stakeholders will be diminished — and/or control over a given object will be questioned. The “outcomes” dimension is the one that confirms, questions, or rejects the effectiveness of the power held by the actor in question. It should be emphasized that power bears upon outcomes, through the status quo, in two ways: maintaining it or altering it.

The emergence of information networks based on microelectronics, of which the global computer network is a part, gave rise to the cyber domain. The “outcomes” dimension is concerned with the incidence of power upon the status quo in the real plane, and not in the digital plane. That is, it is the outcomes that bear upon the real, geographical plane that are considered relevant for the analysis of the diffusion of power, because it is in this plane that the actors are found. Outcomes that bear only upon the digital plane, with no incidence upon the real, exist in a closed system of abstraction and, for this reason, are not relevant for the analysis of the diffusion of power according to the field of IPE. The “outcomes” dimension therefore has, as its principal content, the effect on the geographical reality where the actors are found. Outcomes generated in the cyber plane will be taken into consideration only if they also bear upon geographical reality. In the introduction, we pointed to the example of an individual participating in a virtual game for personal leisure and in the context of a championship.

### 2.7.4 Operationalization of the Dimensions

The three dimensions mentioned above were operationalized so as to become discrete qualitative variables. Each of the three variables receives values that vary according to the dimension, as explained in the introduction of this dissertation.

The first variable, authority, has as its content the sense of trust. This sense concerns the trust that peers and stakeholders in the matter place in the authority in question. The trust deposited (or the lack thereof) designates growth, maintenance, or decline of the actor's level of authority over a given matter. For this reason, this variable can receive the integer values “−1” (decline), “0” (maintenance), and “+1” (growth).

The conceptions of “growth,” “maintenance,” and “decline” were drawn from Strange's own literature (1988, 1996). In *States and Markets*, she discussed this gradual variation of authority. We highlight some examples, such as when she writes about the decline of the authority of the Catholic Church in medieval Christian Europe:

The cultural and social unity provided by the Church created a primitive kind of common market in Europe. It also made possible the accumulation of capital — especially by the great religious orders. As the Church's authority declined, the emerging nation-states inherited from it an economy already pregnant with the growing points of technical change and a commercial structure ready for exploitation by a nascent merchant class. (STRANGE, 1988, p.71–72)

At another point, it is understood that the decline of the Catholic Church's authority within the knowledge structure — that is, with regard to the knowledge produced, stored, and shared in society — was gradually replaced by the growing authority of the scientific State. While the Church saw its power diminish, the scientific State saw its power over society and its activities grow:

[...] It [the Treaty of Westphalia] marked the virtual end of the constraints imposed on kings and princes by the Church. [...] To the same end, the state took over from the Church responsibility for enlarging the education system, in universities as in schools. New patent laws secured monopoly rights for technical innovation through the institution of intellectual property rights. [...] The new technology was also made to serve the interests of the state and to reinforce its power. Even though the technologies of telegraph, railroad and radio were initially developed to serve the interests of business and finance, the cumulative consequence of all three was to tighten the grip of government over the individual. [...] Aided by differences of language, national governments could use technology to keep control by censorship, by monopoly or by restrictive licensing over national systems of education, over national newspapers and broadcasting and even over the publication of books and periodicals. Thus, in this new knowledge structure, the authority of the Church was displaced by the extended authority of the scientific state. [...] In the change, however gradual or slow, from the knowledge structure dominated

by the Church to the knowledge structure dominated by the scientific state, there were certain politically crucial changes. (STRANGE, 1988, p.125–127)

After dominating the knowledge structure and legitimizing itself as the supreme authority within the knowledge structure, the scientific State “fought” for the maintenance of its level of authority — as we can see in this passage highlighted by Strange (1988):

But, once established, the authority of the state, legitimated by the knowledge structure, strove hard to maintain its monopoly position. The more its authority was threatened the more vigorously it was defended. The state, in many cases, asserted a unique right to judge what was acceptable and unacceptable conduct. The Church had asserted its legitimate authority to decide what constituted a ‘state of grace’ rather more than what constituted good conduct. The scientific state asserted its legitimate authority, derived from popular loyalty to and belief in the concept of the nation, to decide what was good conduct, who was loyal or disloyal, what constituted dissidence or treason to the state. (STRANGE, 1988, p.128, emphasis ours).

On another occasion, Strange (1988) explains both the decline of the U.S. government’s authority over U.S.-based transnational corporations and its growth over foreign transnational corporations operating on American territory:

Washington may have lost some of its authority over the US-based transnationals, but their managers still carry US passports, can be sub-poenaed by US courts, and in war or national emergency would obey Washington first. Meanwhile, the US government has gained new authority over a great many foreign corporations operating inside the United States. All of them are acutely aware that the US market is the biggest prize in the competitive game. My guess, from talking to corporate executives — American and European — is that on balance US authority in the world economy has actually increased, not declined. (STRANGE, 1988, p.239).

In *The Retreat of the State*, 1996, in which she discusses the diffusion of power proper, her central argument concerns the decline in the quality of the authority of the governments of most territorial States — and not the decline in the quantity of the exercise of authority (STRANGE, 1996). In any case, her central thesis rests upon the diminution of the authority of States over people and activities within their territories, not the replacement of the State’s authority by other non-state authorities. The existence of other non-state authorities at times competes with, at times rivals, the state authority across the four primary structures defined by Susan Strange’s IPE.

When discussing the rivalry between the authorities of the Italian State and the mafia organization “Cosa Nostra,” Strange (1996) points out that the relationship between them became unstable on two occasions: either when the authority of either party weakened, or when the authority of one of the parties became strong enough to threaten the other. She stated that:

Such arrangements worked (for a time at least) always provided both sides respected — and made their subjects respect — the implicit bargain regarding the division of responsibility. They became unstable however when the authority of either party was weakened, or when the non-state authority became so strong that it was thought to threaten the state. (STRANGE, 1996, p.119, emphasis ours)

Finally, she reinforces the idea of increase, maintenance, and decline of authority when discussing the role of certain professions in the global political economy landscape, such as insurance business managers:

Another consequence is to give added authority to the insurers. As technological and financial change affects their business, they respond by putting a higher price on premiums for some risks over others, or by refusing to offer insurance cover on any terms whatsoever. [...] They can, and indeed do, exercise the same kind of arbitrary authority over others when they refuse, for example, to sell insurance against theft to homeowners unwise — or unlucky — enough to live in burglary prone streets. Or, when they deny marine insurance to shipowners whose masters take the ship into a war-zone, as happened during the Iraq war. [...] For if, as I have argued, authority in political economy is recognisable by the power to alter or modify the behaviour of others by using incentives and disincentives to affect the choice and range of options, there can be little doubt that as the world economy grows, the costs and risks of economic transactions escalate, allowing insurers and reinsurers to exercise increasing authority in and over the system. (STRANGE, 1996, p.133)

Given that Strange (1988, 1996), on various occasions, discussed authority and fluctuations of authority, we believe that “growth,” “maintenance,” and “decline” of authority bear upon the balance of power.<sup>63</sup> The “authority” variable represents one of the dimensions of the diffusion of power.

The second variable, control, has as its content the sense of exercise or operation of a given object by an actor. Control over objects can be absolute, partial, or none. It is

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<sup>63</sup>The balance of power is understood here as the interplay of forces, of power among state and non-state actors in the IS. It should not be understood in the classical sense of the expression as used in Realism.



absolute when the actor in question is solely responsible for the control of a given object, directing it as they see fit. It is partial when they are not solely responsible for controlling the object, or when they depend on other actors to control the object. And there is no control when they are not responsible for controlling the object, or when they have no access to the object.

A classic example of the role of control in Strange's literature (1988) concerns the power that bureaucracy has over the object produced by producers and consumed by consumers — which directly affects the production and the economy of a domestic market, for example. She stated that:

Even in a command economy, there is, behind the veil of bureaucratic control, a kind of bargain between authority in the form of state ministries, and market in the form of consumers and producers. To maintain the authority of the state, a bargain has to be struck with the producers — managers and workers — to reward them sufficiently and to give effective enough incentives for them to produce the goods and services that will sell to consumers. (STRANGE, 1988, p.39–40)

The incentives that the State provides to producers, so that they produce a sufficient quantity to meet the consumer market, must be such that the absence of the product does not bear upon the economy, for, after all, the producers have absolute control over the product — not the State. The latter, indirectly, makes use of incentives to bear upon production. Its control over the product is partial.

Another example discusses multinational companies that suffer appropriation by the State. Although these companies are compensated, as mandated by international law, the State often faces frustrations: in such cases, the States hold the production, the machinery, the resources to exploit the market, but not the technology, knowledge, and market access. After all, market access remains under the control of the former dispossessed companies. It is they who hold control over technology, knowledge, and market access. And this control means power. In particular, Strange (1988) cites the cases of Nigeria and Chile, stating that these countries:

They began by nationalizing them, first the mineral and oil companies and then banks, insurance, breweries and other enterprises. Their right to do so — since industrialized countries had often done the same — was unchallenged provided only that they observed the rule of customary international law that compensation should be made promptly, in full and equitably. Yet the developing country governments very often found that they had won an empty victory, and too often at a high price. They had the mines, or the oil wells, but not the same power to exploit the market. Whether it was Chilean copper

or the Guinness brewery in Nigeria, the displaced companies kept control over market access, by making long-term contracts with the customers, for instance. They also had command of the technology necessary to remain competitive in world markets. (STRANGE, 1988, p.85, emphasis ours)

In the case of the knowledge structure, the Catholic Church had its power and authority reinforced by control over the means of communication (sacred books, literacy in the Latin language — official within religious rites) (STRANGE, 1988). For Strange (1988), it is control over access to knowledge that enabled the maintenance of power within the Catholic Church — which indicates that rival authorities within this structure had to be eliminated or discredited (STRANGE, 1988, p.124). The Catholic Church's control in medieval Christian Europe over knowledge was, at times, absolute. The weakening of the Church's authority within the knowledge structure, in the face of the growing authority of the scientific State, brought to the fore the relevance of science and the scientific community. The axis of supreme authority within the knowledge structure shifted from the Church to the State. The sacred scriptures and the Latin language were replaced by scientism and the English language. The belief, on the part of peers and stakeholders (society in general and national governments, principally), in Science (and in the scientific community, consequently) as an authority over natural facts signified the erosion of the Church's authority over the same matter — relegating it to questions of spirituality and human faith (STRANGE, 1988). In this context, the Church's control over the means of communication did not retain the same relevance: postal services, telegraphs, and the telephone — originating from discoveries by the scientific community and controlled by States and private companies — meant power for those who held their control. In other words, the objects relevant for the achievement of desired outcomes within the knowledge structure changed, and they meant power for whoever held control over these new objects.

On another occasion, Strange (1996) cites the monopoly of control by companies over a technology, supply, marketing system, or even a brand. In the global market, this monopoly means power over the set of activities of the sector and consumers interested in the specific products of this company. Over time, it may be that the technology becomes shared (reverse engineering is one of the most commonly used tools) and the power that derives from control is, little by little, eroded — until a new technology is developed and monopoly over it is established. Or, as in the case of cartels that control the world's supply of diamonds: their power derives from the tight, nearly absolute, control over supplies. According to Strange:

[...] She admits that the most effective cartel of the four she studied, that in diamonds, almost entirely owed its success to the tight control over supplies

exercised by one firm, Anglo-American, and the majority owners, the Oppenheimer family. The supporting role of the South African, Soviet and Israeli governments was just that — supportive. (STRANGE, 1996, p.149).

There are other examples of control over an object, whether total, partial, or none. We suggest consulting “Appendix 3 — Control, 1988” and “Appendix 4 — Control, 1996.”

Finally, regarding the “outcomes” variable. There are only two types of outcomes: those that alter the status quo; and those that do not. The alteration of the status quo (of the current state of affairs, or rather, the present state of something) signifies a change of state. Altering the status quo entails the replacement of one situation by another. The non-alteration of the status quo, in turn, entails its permanence, its reinforcement. In other words, there is a contest between forces for the alteration or permanence of the status quo. There is no alternative beyond these two. It should merely be recalled that the status quo refers to the region where the actors are found, that is, the real, geographical plane (and not the abstraction of information systems such as the cyber domain).